# Introduction to Brain and Cognition

## Topic: Attention Perception and Performance

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# Attention, Perception, & Performance

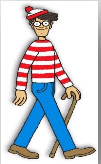
Wally or Waldo – Where's Wally, Children puzzle book, by Martin Hanford, 1987

Activity - 1

What am I seeing?

Bottom-up processing:

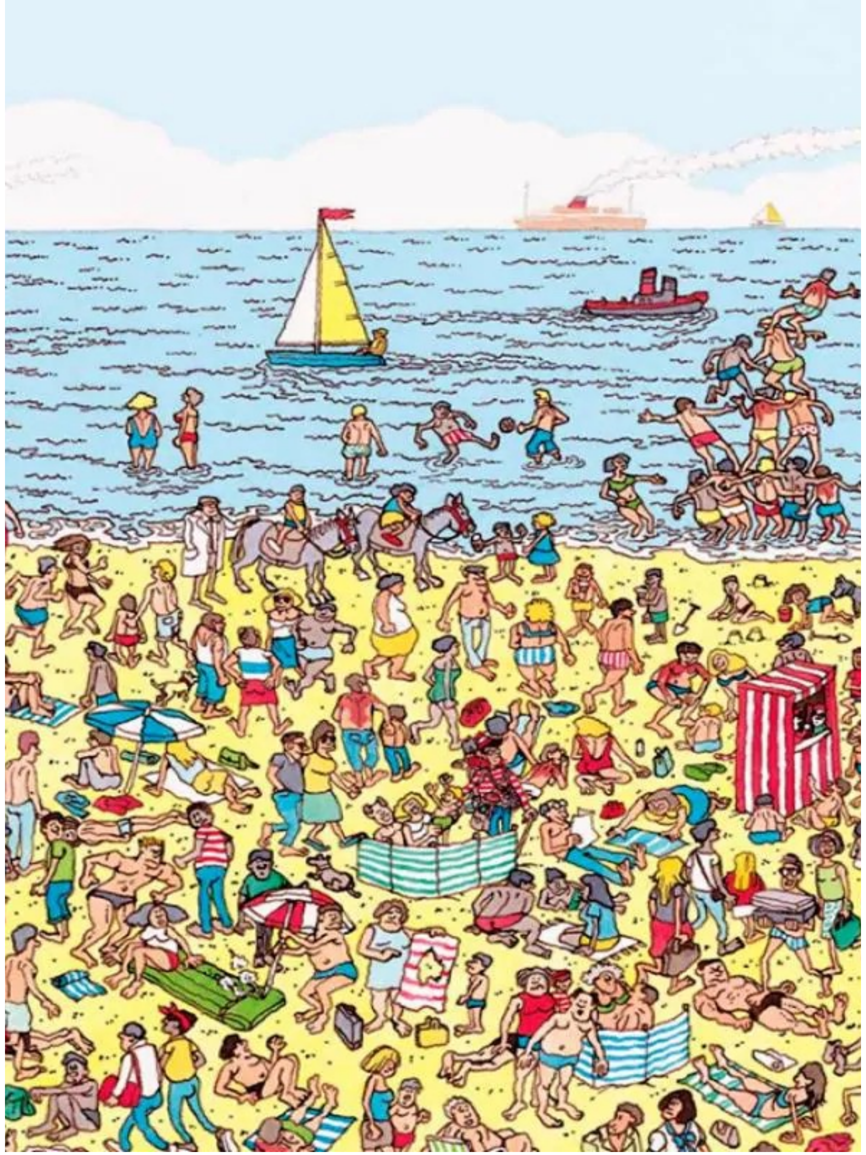
taking sensory information and then assembling and integrating it



Top-down processing:

using models, ideas, and expectations to interpret sensory information

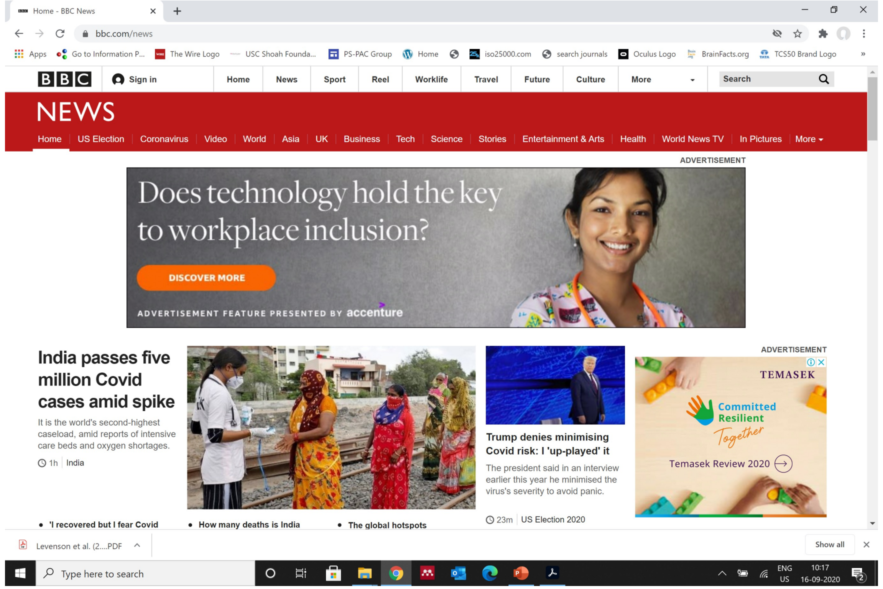
Is that something I've seen before?

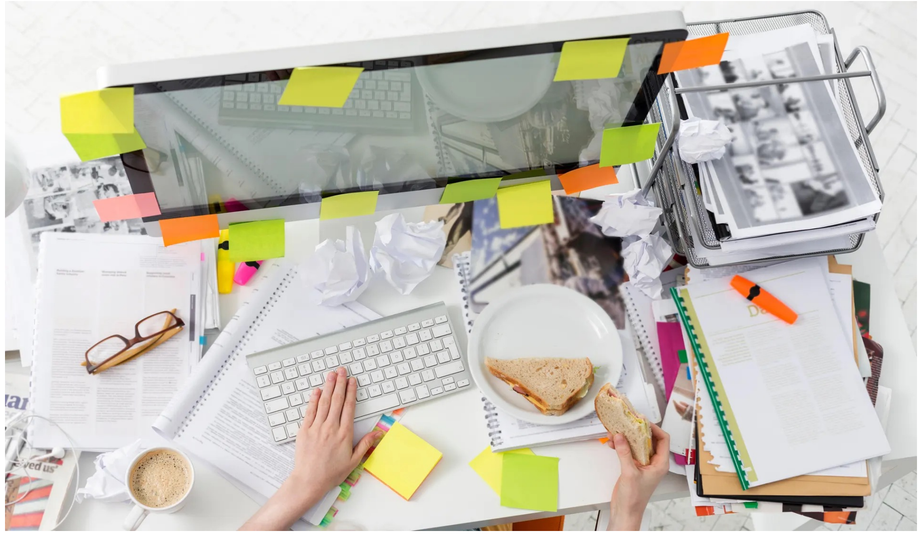


Where’s Wally in Real Time !

The real complex conjunctive search

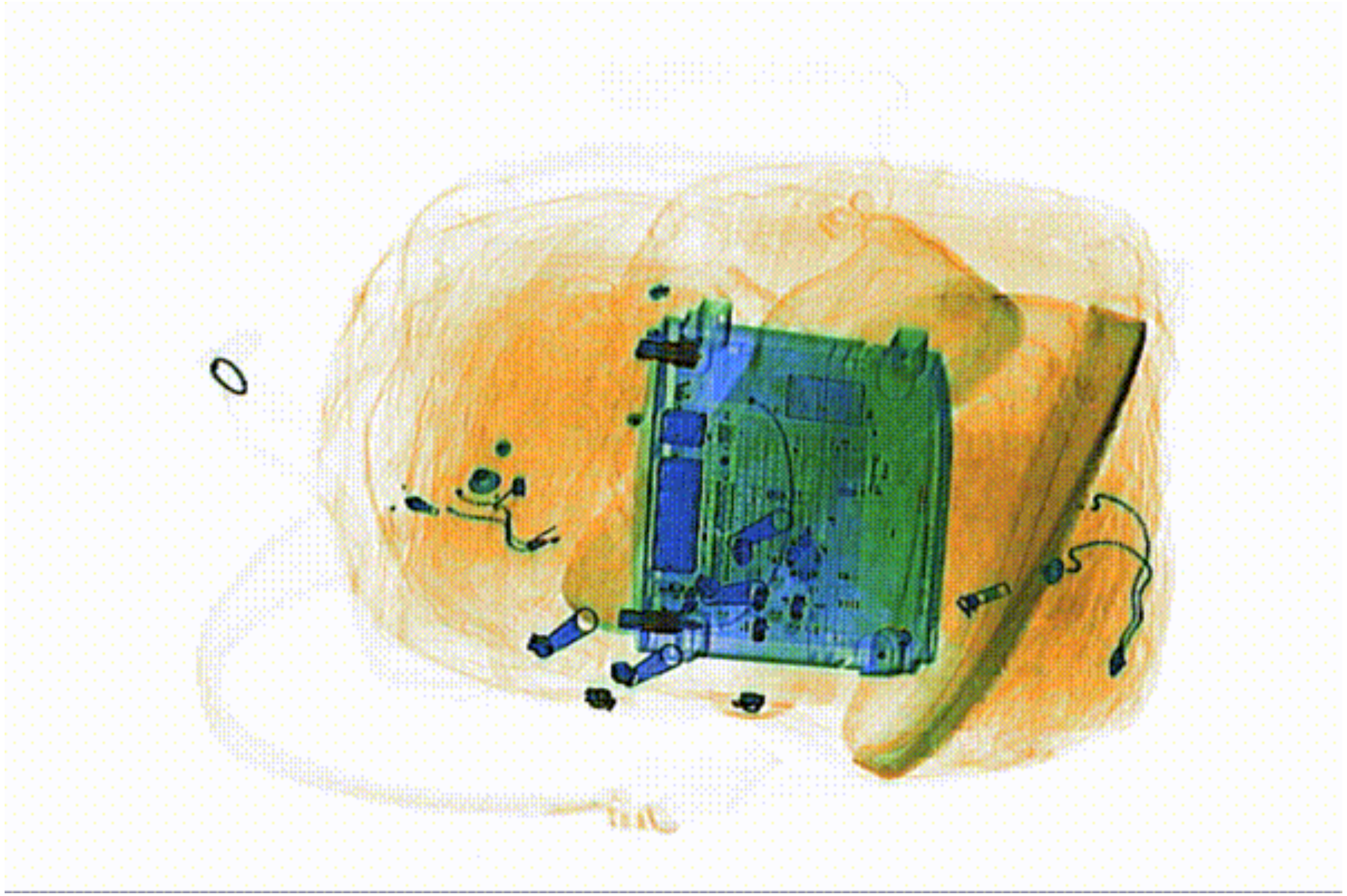
# Where's Wally in Real Time? Visual search task



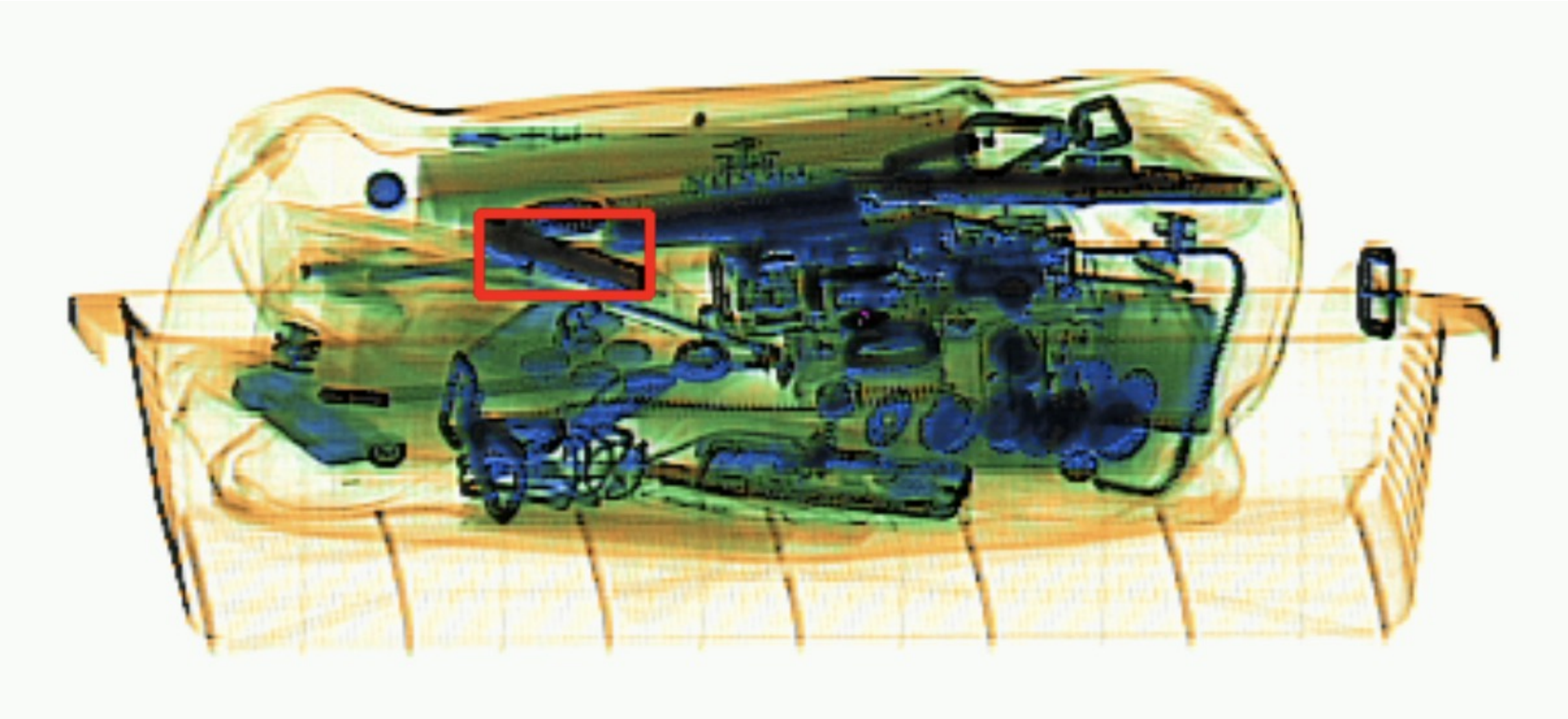
# Let’s see how scanning happens here?

# How about an Airport security check – detecting threat items in luggage
# Self-terminative *vs.* Exhaustive Search



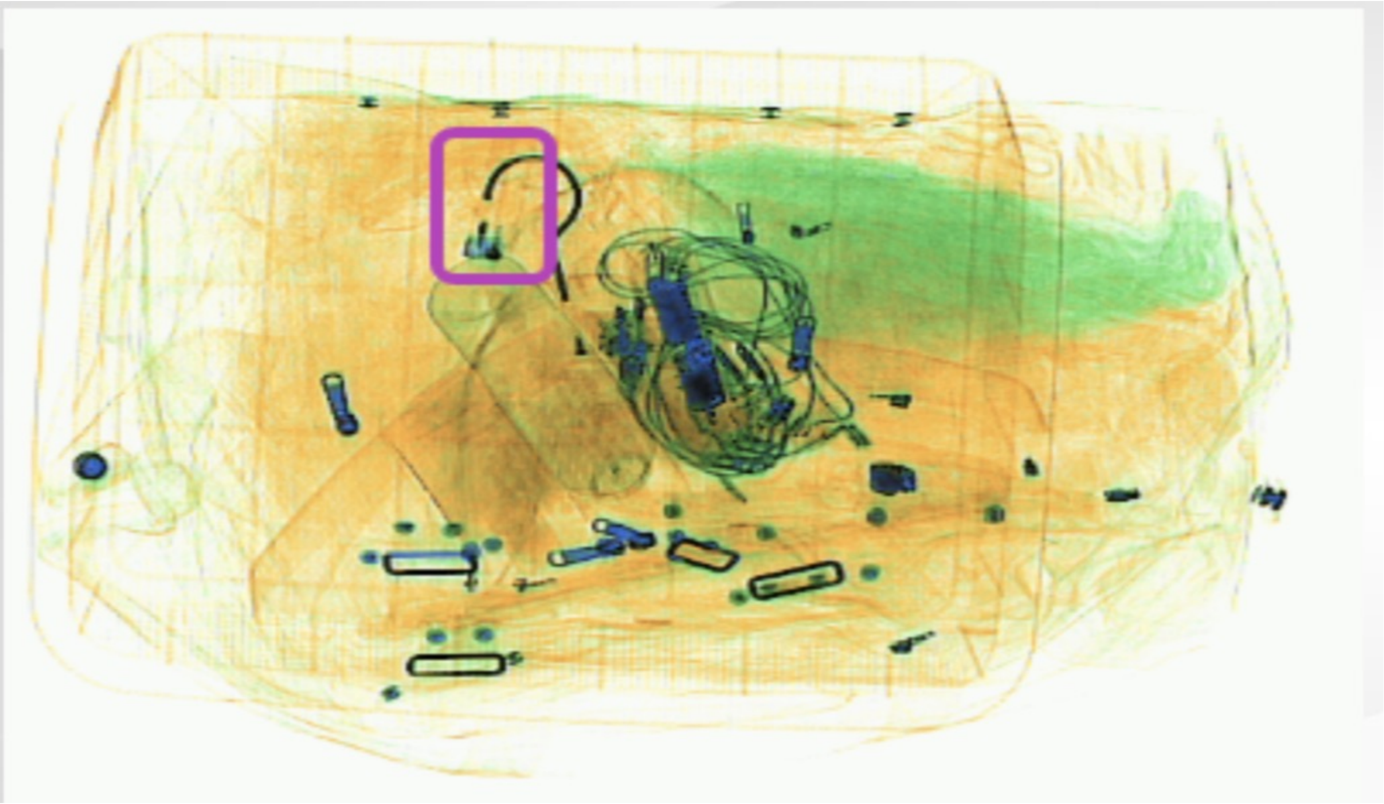
- Sample set of 30 X-rays of baggage. This set contains at least two firearms and three knives.
- Exhaustive search (active search until the entire set is exhausted or searched)

Contd.



The “side view” of a very cluttered bag with several dense metallic objects that appear as black or dark blue, including coins, a buckle, a tablet. The tray and bag, low-density and made of organic material, appear as a faint orange. A pocket knife is shown in the red box, which would be very difficult to spot with only a “top” view.

# Contd.



Some items can blend in much more easily with their surroundings. Pictured is a bag containing a lighter. In Japanese airports, passengers are not allowed to carry more than one lighter onto a plane

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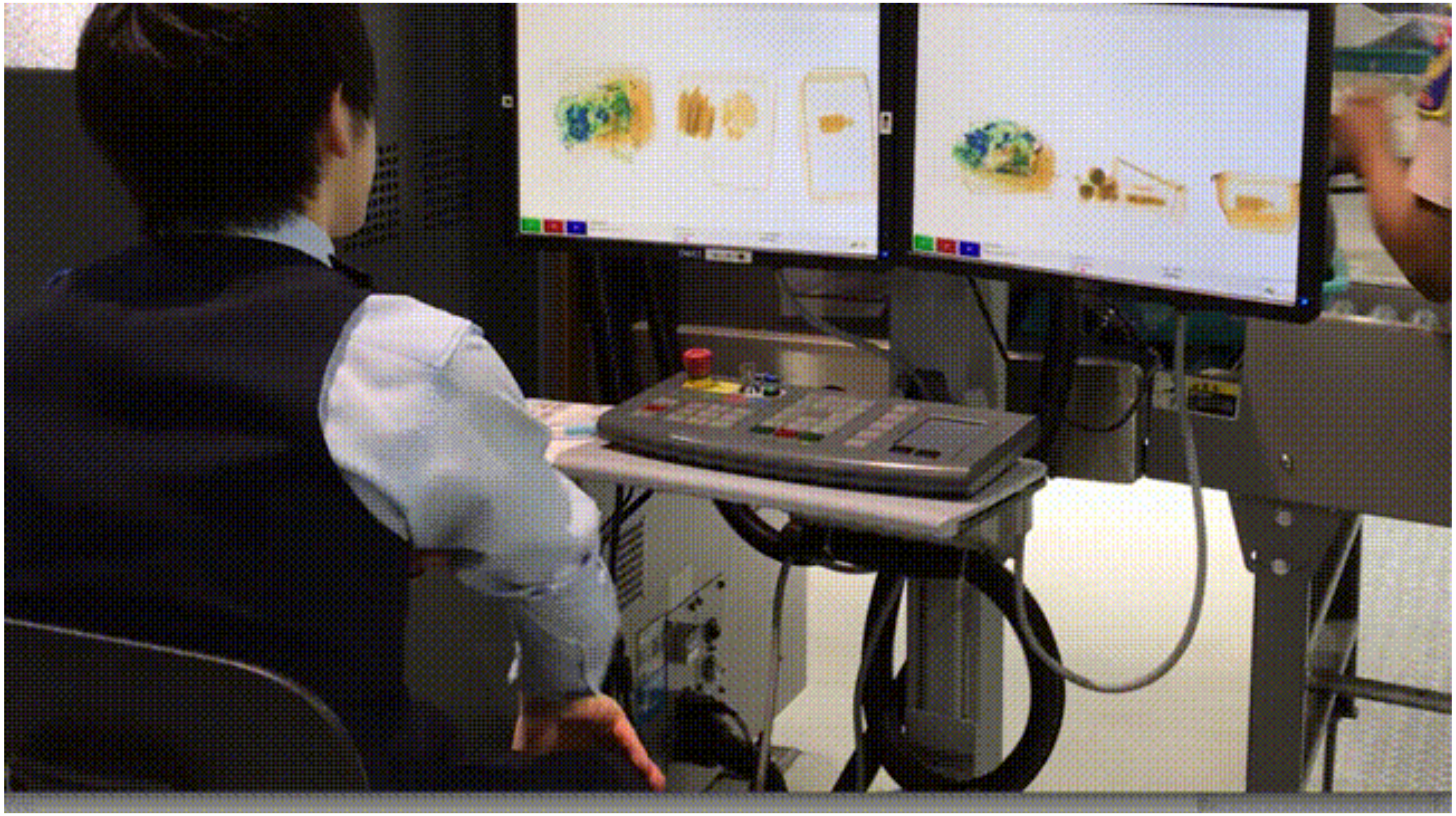

An X-ray security operator screening passengers

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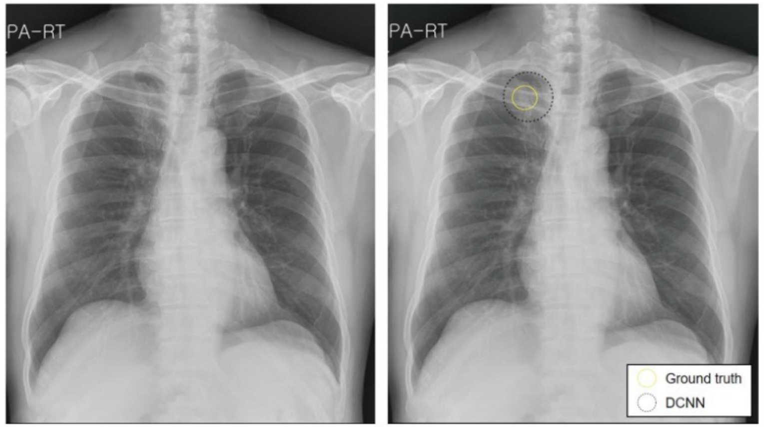
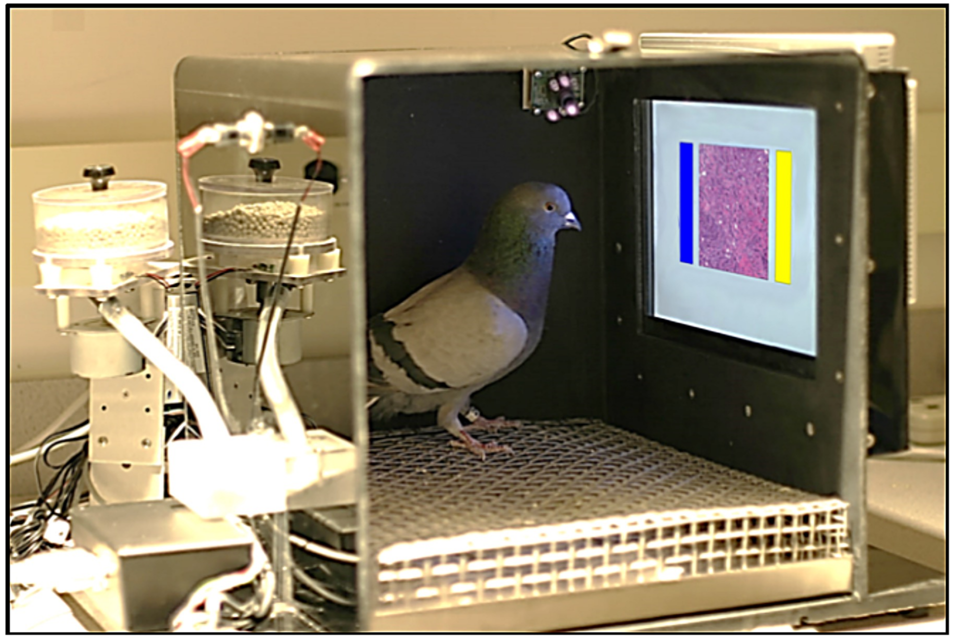



Radiologist screening image for cancer detection in simple versus conjunctive search problem.



## Pigeons (*Columba livia*) as Trainable Observers of Pathology and Radiology Breast Cancer Images

**Richard M. Levenson[1]*, Elizabeth A. Krupinski[3], Victor M. Navarro[2], Edward A. Wasserman[2]***

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OPEN ACCESS

**Citation:** Levenson RM, Krupinski EA, Navarro VM, Wasserman EA (2015) Pigeons (*Columba livia*) as Trainable Observers of Pathology and Radiology Breast Cancer Images. PLoS ONE 10(11): e0141357. doi:10.1371/journal.pone.0141357

**Editor:** Jonathan A Coles, Glasgow University, UNITED KINGDOM

## Abstract

Pathologists and radiologists spend years acquiring and refining their medically essential visual skills, so it is of considerable interest to understand how this process actually unfolds and what image features and properties are critical for accurate diagnostic performance. Key insights into human behavioral tasks can often be obtained by using appropriate animal models. We report here that pigeons (*Columba livia*)—which share many visual system properties with humans—can serve as promising surrogate observers of medical images, a capability not previously documented. The birds proved to have a remarkable ability to distinguish benign from malignant human breast histopathology after training with differential food reinforcement; even more importantly, the pigeons were able to generalize what they had learned when confronted with novel image sets. The birds' histological accuracy, like that of humans, was modestly affected by the presence or absence of color as well as by degrees of image compression, but these impacts could be ameliorated with further training.

# Attention Attention


- What is attention? Why is it important?
- How does it influence perception and other cognitive processes?
- What directs our attention? How do we decide what to attend to?
- Where does attention take place?


What is Attention ?
Why is it important for Cognitive Systems ?

# Activity 2 - Let’s watch a video

**IN FOLLOWING VIDEO – COUNT HOW MANY TIMES PLAYERS WEARING WHITE T-SHIRT PASS THE BALL**

# Wondering how did you FAIL to Notice Gorilla or the other changes?

Task demand

Wasn't familiar with the task

Wasn't Expecting

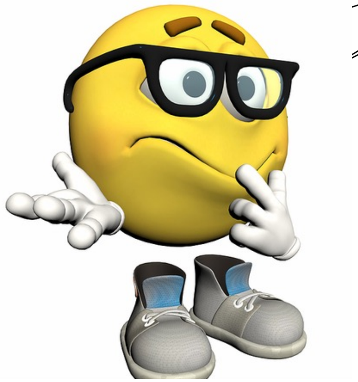
Had/ having bad day/ stress / wasn't feeling well

Mind wandering

Too many and too rapid

Wasn't interesting or else

Got distracted



**Task demand, Cognitive Load,** but Stress, Fatigue, Motivation, Interest, Knowledge, Belief, Emotion, and Feeling also modulates the filter and central attention

Wondering how did you FAIL to Notice Gorilla or the other changes?

Wasn't familiar with the t

Task demand

many and too rapid

Got distracted

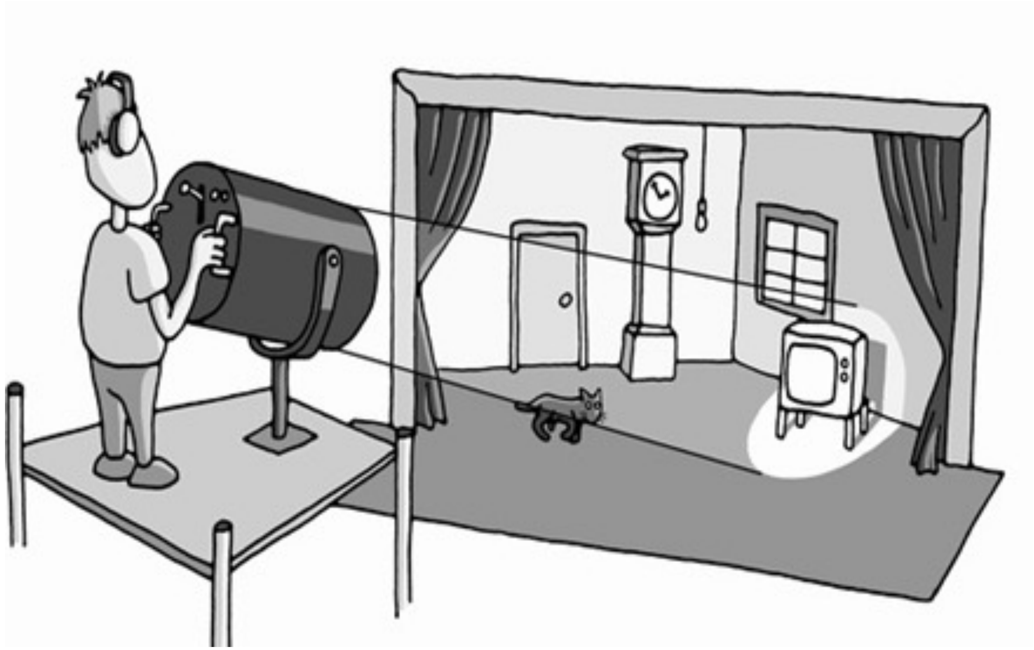
Wasn't iteresting or else

Attention is LIMITED !

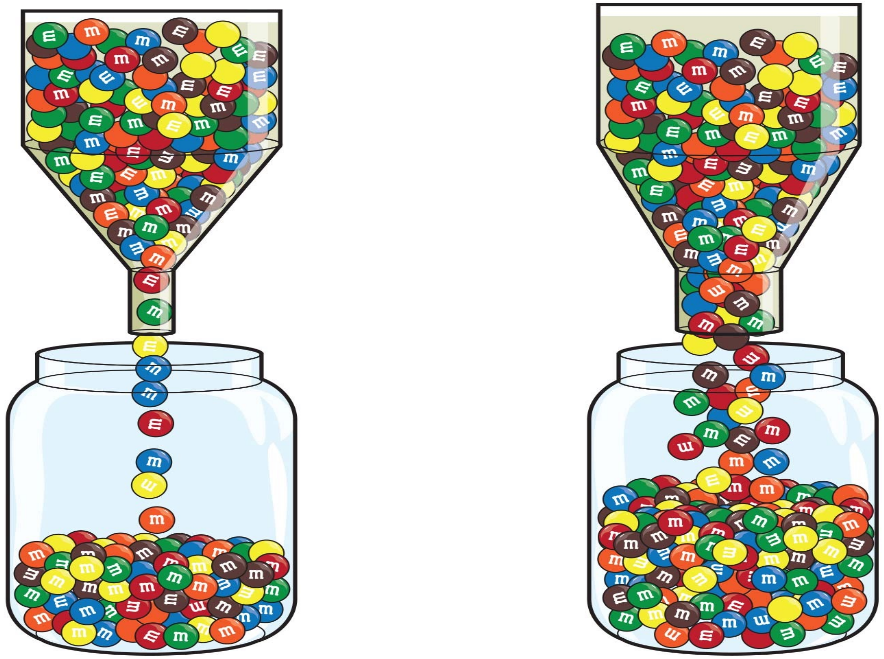
**Task demand, Cognitive Load,** but Stress, Fatigue, Motivation, Interest, Knowledge, Belief, Emotion, and Feeling also modulates the filter and central attention

# What is Attention?

Selection, Filter, and Focus



**Selective Focus - SPOTLIGHT**



**Filter and Bottleneck - GATEKEEPER**

Enable **focus** on **relevant information selectively while ignoring irrelevant information**

**What is Attention?**

Shifting, Alternating, and Dividing – Dual Task

**Central Attention / Resource**

- Enable dividing your attention to more than one task - multi-tasking / dual task
- Do we have single or multiple attention resources? (We have multiple resources to juggle between tasks)
- The efficiency of juggling between task depends on practice, task complexity, stimulus features, task demand, sharing of stimulus features / actions between two tasks

# How Does This Limitation Affect Our Performance?

Let’s look at

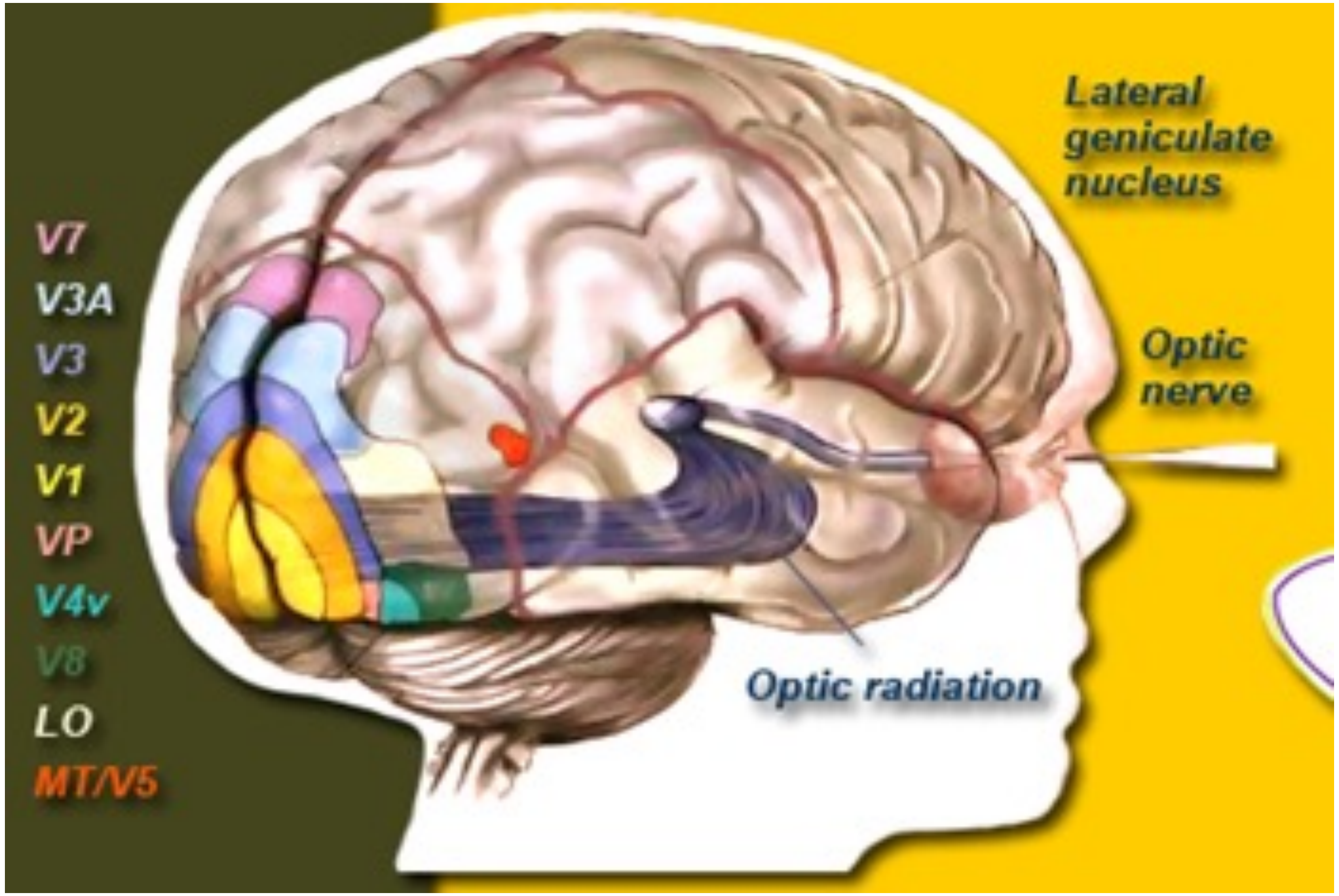
# Why attention is important ?

## Influence Perception

- **Inattention Blindness – Failure to perceive the object that are not the focus of attention**


- **Change Blindness – failure to detect changes to the visual details of a scene**

## Contd.





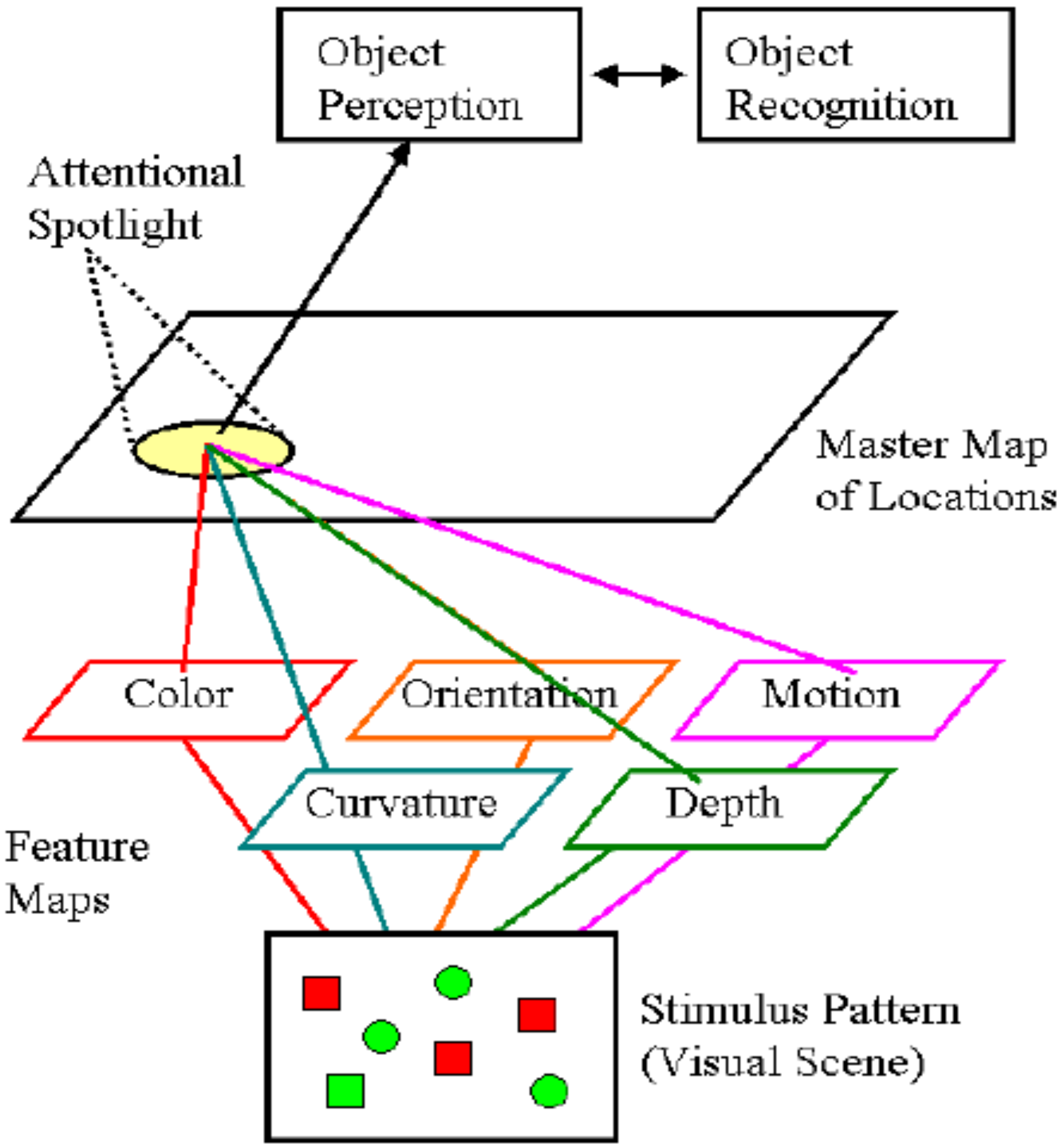

**BINDING Problem**

How does you brain combines color, shape, motion, and location into a single object.

In other words, how your brain binds these various features together which are present in this complex dynamic scene?

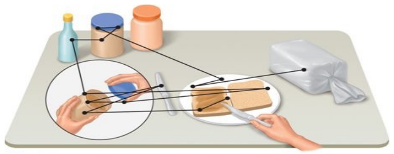
# Contd.

Feature Integration Theory (Treisman)

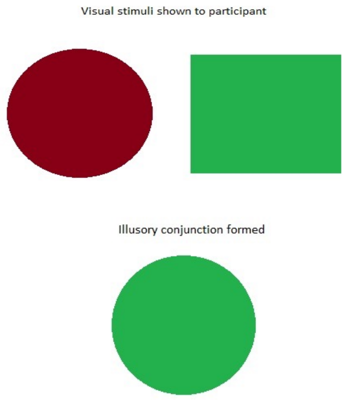



Attention work as a GLUE, Treisman & Gelade, 1980

## When attention fails



BINDING PROBLEM




Illusory Conjunction

# Why attention is important ?

**ACTIVITY 3 - Read**

| | | | |
|---|---|---|---|
| X | M | L | T |
| A | F | N | B |
| C | D | Z | P |

**ACTIVITY 3 - Recall**

RECALL

**ACTIVITY 3 - Read**

| | | | |
|---|---|---|---|
| J | T | L | X |
| D | G | I | Q |
| Y | W | R | S |

# ACTIVITY 3 - Recall

LEFT

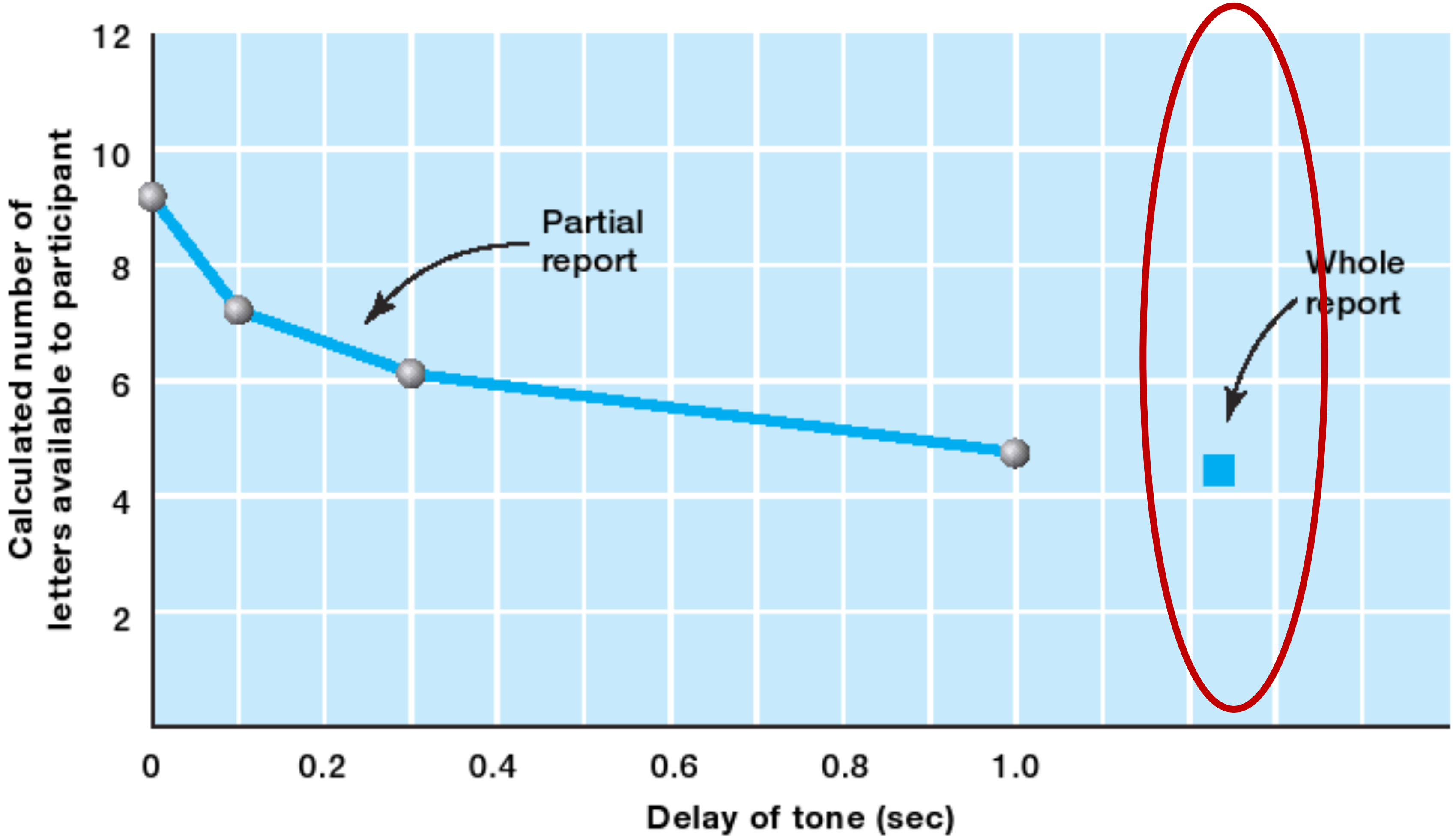
MIDDLE

RIGHT

# Why attention is important ?

## Influence Memory


Figure 5.6 Procedure for three of Sperling's (1960) experiments. (a) Whole report procedure: Person saw all 12 letters at once for 50 msec. and reported as many as he or she could remember. (b) Partial report: Person saw all 12 letters, as before, but immediately after they were turned off, a tone indicated with row the person was to report; (c) Partial report, delayed: Same as (b), but with a short delay between extinguishing the letters and presentation of the tone.




**Sperling Experiment**

Contd.

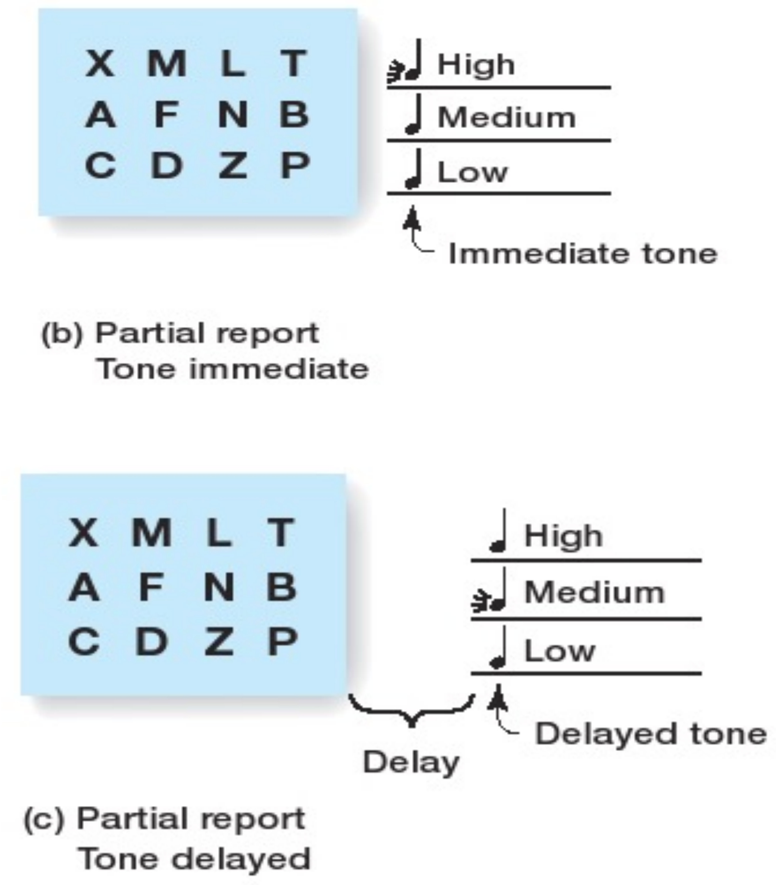
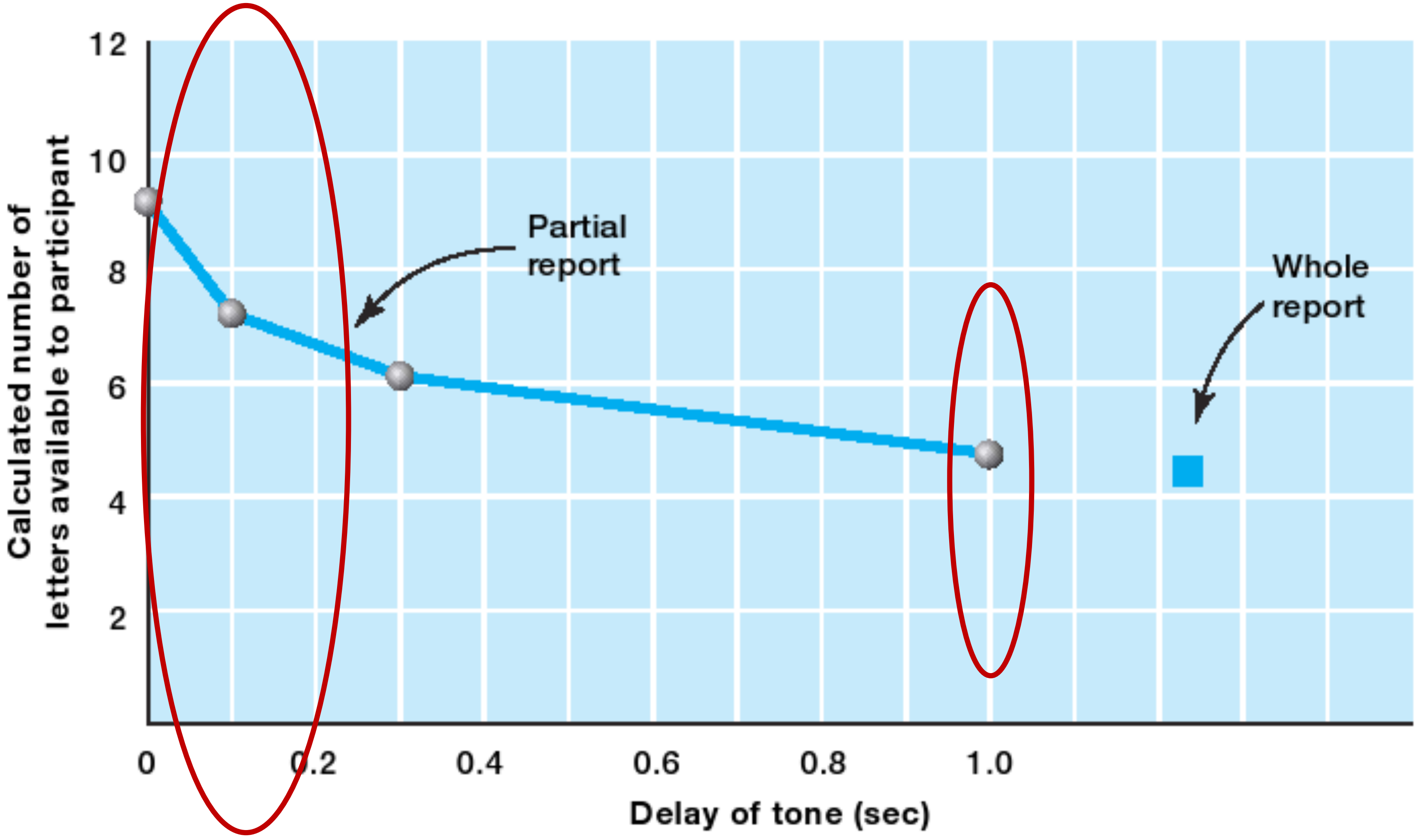



Figure 5.7 Results of Sperling's (1960) partial-report experiments. The decrease in performance is due to the rapid decay of iconic memory (called *sensory memory* in the modal model). (Reprinted from "The Serial Position Effect in Free Recall," by B. B. Murdoch, *Journal of Experimental Psychology, 64*, pp. 482–488. Copyright © 1962 with permission from the American Psychological Association.)

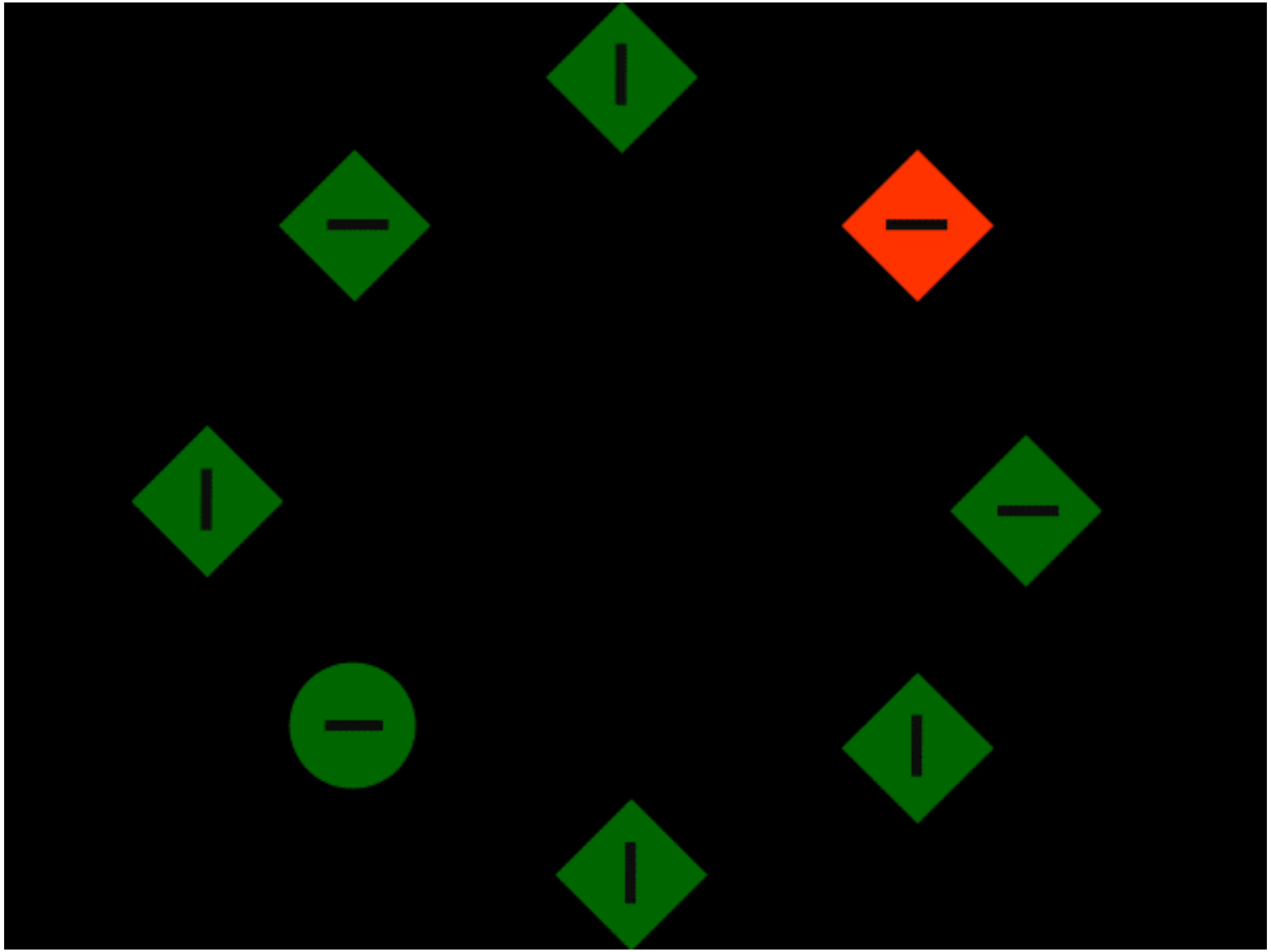



**Sperling Experiment**

# What Directs Our Attention?

# What directs our attention ?

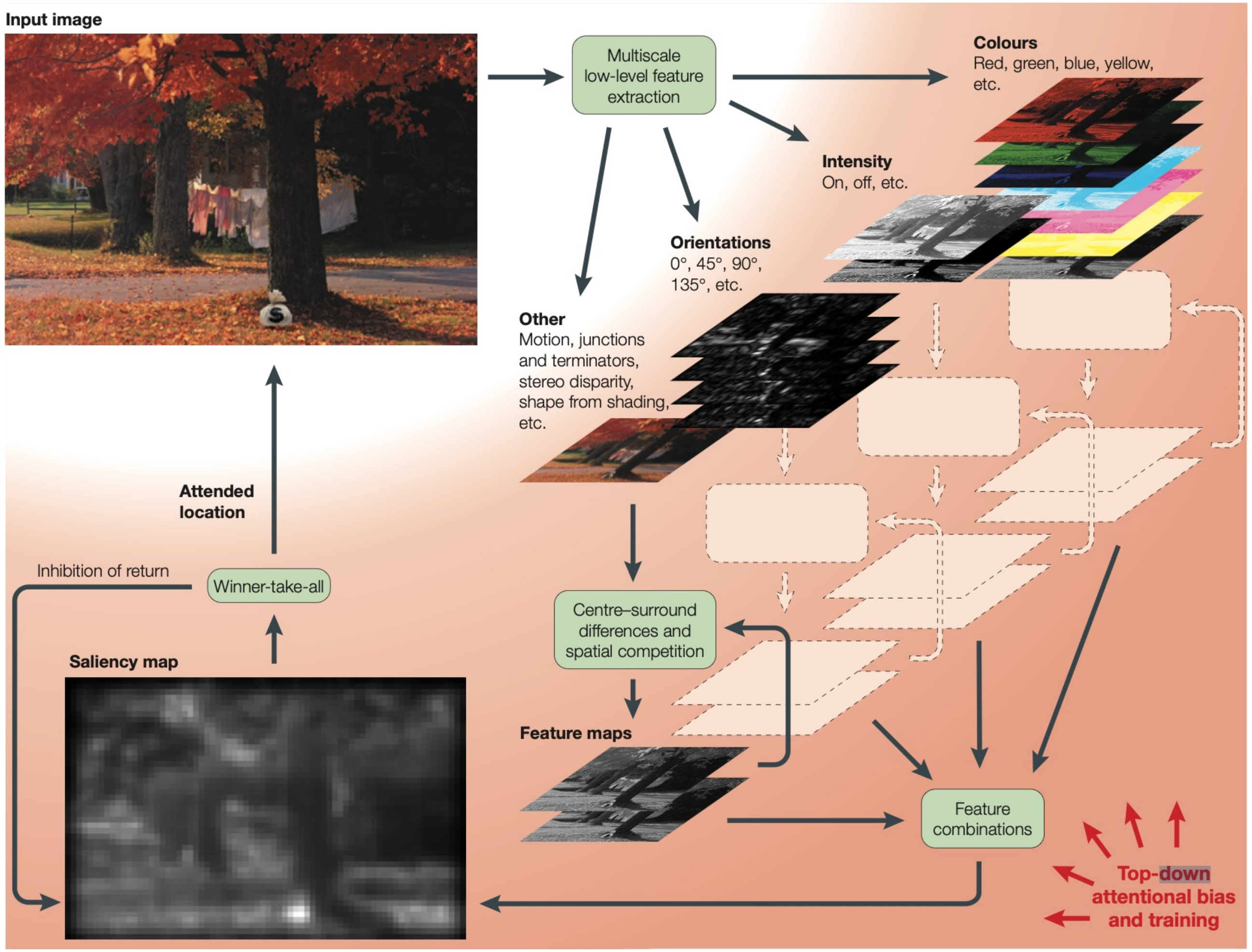
Saliency, Abrupt change

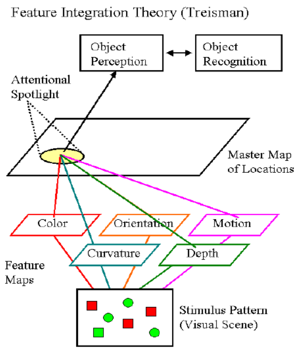


An example of attention capture: When participants were instructed to search for green circle they often looked at the orange/ red diamond first.

# Contd.

REVIEWS




REVIEWS

# COMPUTATIONAL MODELLING OF VISUAL ATTENTION


*Laurent Itti* and Christof Koch‡*



Five important trends have emerged from recent work on computational models of focal visual attention that emphasize the bottom-up, image-based control of attentional deployment. First, the perceptual saliency of stimuli critically depends on the surrounding context. Second, a unique 'saliency map' that topographically encodes for stimulus conspicuity over the visual scene has proved to be an efficient and plausible bottom-up control strategy. Third, inhibition of return, the process by which the currently attended location is prevented from being attended again, is a crucial element of attentional deployment. Fourth, attention and eye movements tightly interplay, posing computational challenges with respect to the coordinate system used to control attention. And last, scene understanding and object recognition strongly constrain the selection of attended locations. Insights from these five key areas provide a framework for a computational and neurobiological understanding of visual attention.


CENTRE–SURROUND MECHANISMS
These involve neurons that respond to image differences between a small central region and a broader concentric antagonistic surround region.

The most important function of selective visual attention is to direct our gaze rapidly towards objects of interest in our visual environment[1–9]. This ability to orientate rapidly towards salient objects in a cluttered visual scene has evolutionary significance because it allows the organism to detect quickly possible prey, mates or predators in the visual world. A two-component framework for attentional deployment has recently emerged, although the idea dates back to William James[1], the father of American psychology. This framework suggests that subjects selectively direct attention to objects in a scene using both bottom-up, image-based saliency cues and top-down, task-dependent cues.

Some stimuli are intrinsically conspicuous or salient in a given context. For example, a red dinner jacket among black tuxedos at a sombre state affair, or a flickering light in an otherwise static scene, automatically and involuntarily attract attention. Saliency is independent of the nature of the particular task, operates very rapidly and is primarily driven in a bottom-up manner, although it can be influenced by contextual, salient, it will pop out of a visual scene. This suggests that saliency is computed in a pre-attentive manner across the entire visual field, most probably in terms of hierarchical CENTRE–SURROUND MECHANISMS. The speed of this saliency-based form of attention is on the order of 25 to 50 ms per item.

The second form of attention is a more deliberate and powerful one that has variable selection criteria, depending on the task at hand (for example, 'look for the red, horizontal target'). The expression of this top-down attention is most probably controlled from higher areas, including the frontal lobes, which connect back into visual cortex and early visual areas. Such volitional deployment of attention has a price, because the amount of time that it takes — 200 ms or more — rivals that needed to move the eyes. So, whereas certain features in the visual world automatically attract attention and are experienced as 'visually salient', directing attention to other locations or objects requires voluntary 'effort'. Both mechanisms can operate in parallel.

Attention implements an information-processing bottleneck that allows only a small part of the incoming


**Hedco Neuroscience Building, University of Southern California, 3641 Watt Way, Los Angeles, California 90089-2520, USA. ‡Computation and Neural Systems Program, Caltech, Pasadena, California 91125, USA. Correspondence to L.I.*

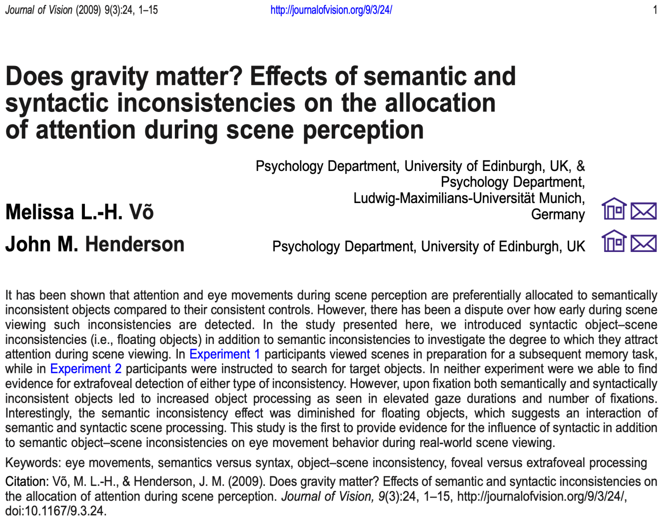
# What directs our attention ?

## Knowledge / Schema

Semantically consistent | Semantically inconsistent

Syntactically consistent

Syntactically inconsistent

Journal of Vision (2009) 9(3):24, 1–15 http://journalofvision.org/9/3/24/ 1

### Does gravity matter? Effects of semantic and syntactic inconsistencies on the allocation of attention during scene perception

Melissa L.-H. Võ — Psychology Department, University of Edinburgh, UK, & Psychology Department, Ludwig-Maximilians-Universität Munich, Germany

John M. Henderson — Psychology Department, University of Edinburgh, UK

It has been shown that attention and eye movements during scene perception are preferentially allocated to semantically inconsistent objects compared to their consistent controls. However, there has been a dispute over how early during scene viewing such inconsistencies are detected. In the study presented here, we introduced syntactic object–scene inconsistencies (i.e., floating objects) in addition to semantic inconsistencies to investigate the degree to which they attract attention during scene viewing. In Experiment 1 participants viewed scenes in preparation for a subsequent memory task, while in Experiment 2 participants were instructed to search for target objects. In neither experiment were we able to find evidence for extrafoveal detection of either type of inconsistency. However, upon fixation both semantically and syntactically inconsistent objects led to increased object processing as seen in elevated gaze durations and number of fixations. Interestingly, the semantic inconsistency effect was diminished for floating objects, which suggests an interaction of semantic and syntactic scene processing. This study is the first to provide evidence for the influence of syntactic in addition to semantic object–scene inconsistencies on eye movement behavior during real-world scene viewing.

Keywords: eye movements, semantics versus syntax, object–scene inconsistency, foveal versus extrafoveal processing

Citation: Võ, M. L.-H., & Henderson, J. M. (2009). Does gravity matter? Effects of semantic and syntactic inconsistencies on the allocation of attention during scene perception. *Journal of Vision, 9*(3):24, 1–15, http://journalofvision.org/9/3/24/, doi:10.1167/9.3.24.

*Vo and Henderson, 2009*

# What directs our attention ?

## Knowledge / Schema


- Semantically inconsistent objects were fixated (total fixation) on for significantly longer than semantically inconsistent objects, $F(1,23) = 6.36$, $p < .05$.
- Floating objects, i.e., syntactically inconsistent, were looked at significantly longer than objects resting on surfaces, $F(1,23) = 42.43$, $p < .01$.
- However, the interaction failed to reach significance, $F(1,23) = 1.94$, $p > .05$.
- Similar to total fixation duration, semantically and syntactically inconsistent objects led to a significantly greater number/count of fixations than semantically consistent ($F(1,23) = 7.07$, $p = .01$) and syntactically consistent ($F(1,23) = 29.91$, $p < .01$) objects.

*Vo and Henderson, 2009*

# What directs our attention ?

Goal / Task demand

Ilya Repin, *An Unexpected Visitor*, (1884)

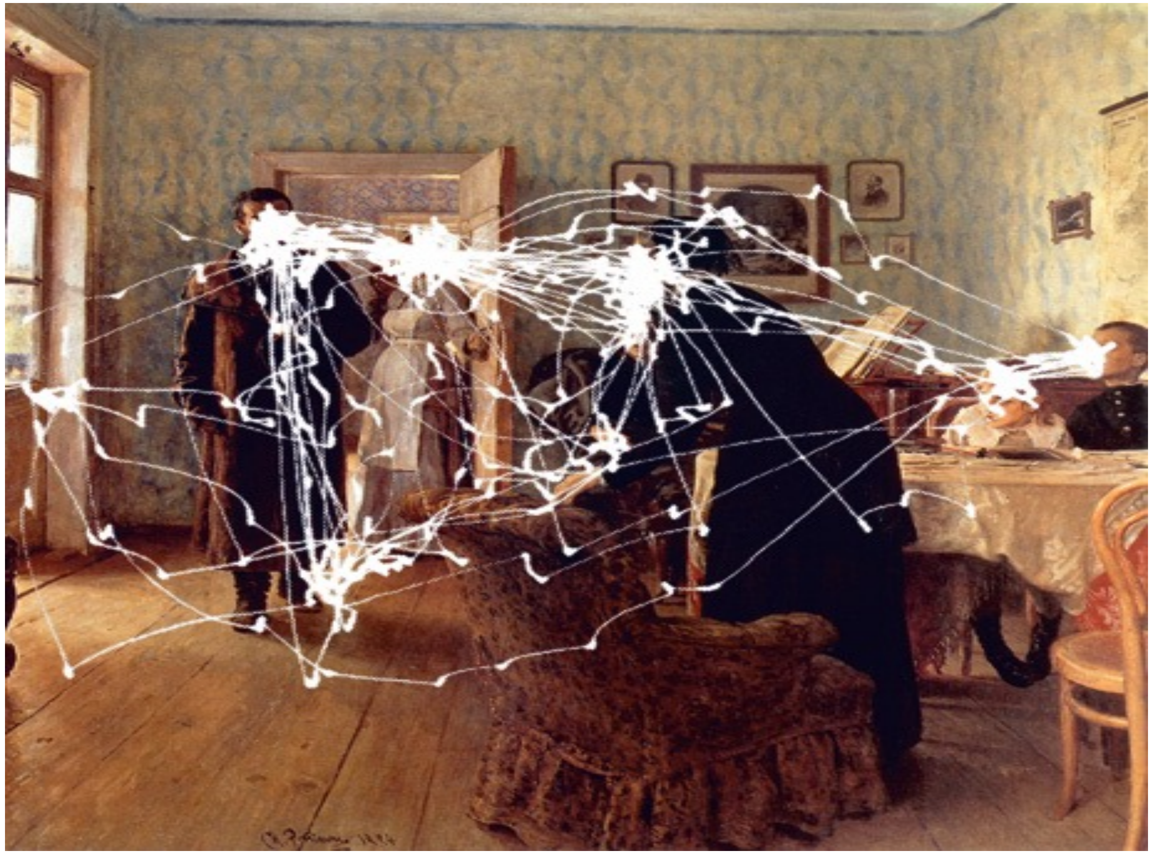
- Participants were asked to look at the painting with different questions in mind.

- The same painting was viewed differently, when asked to attend to different components of the painting.

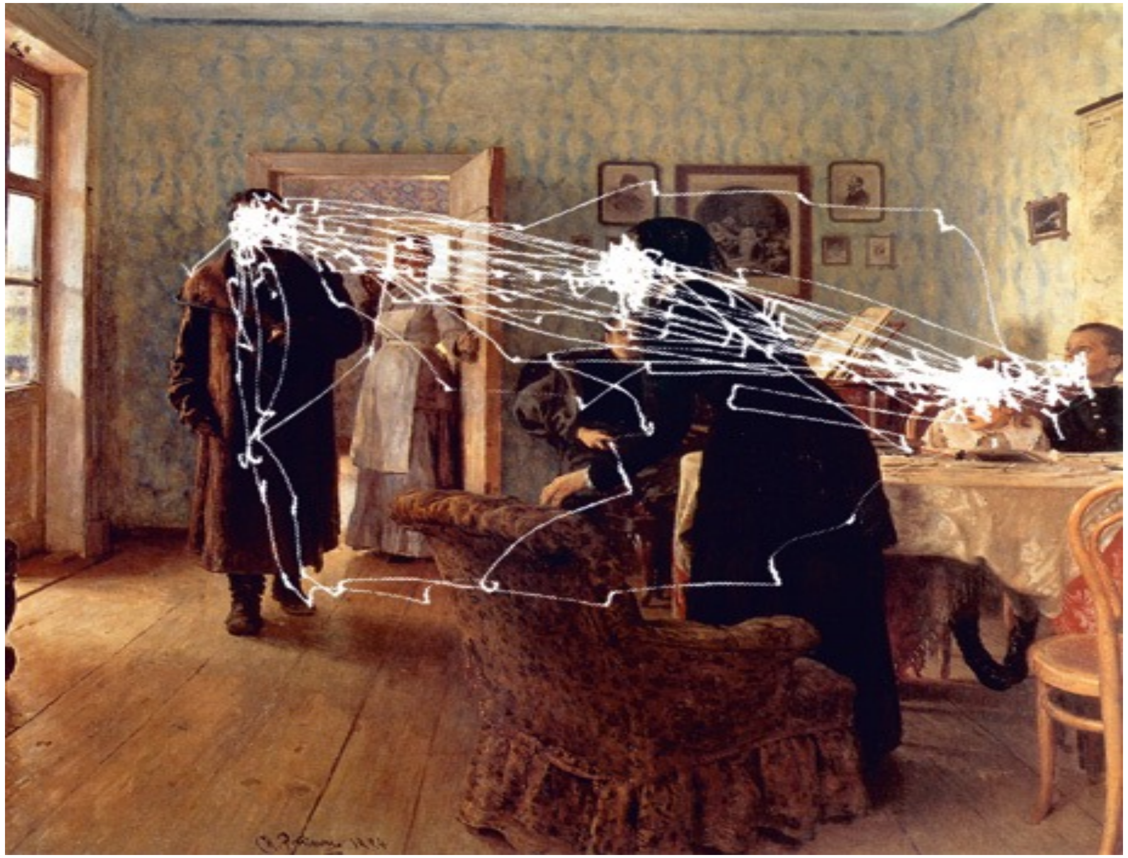
Yarbus, 1967

# What directs our attention ?

Goal / Task demand



Free Examination

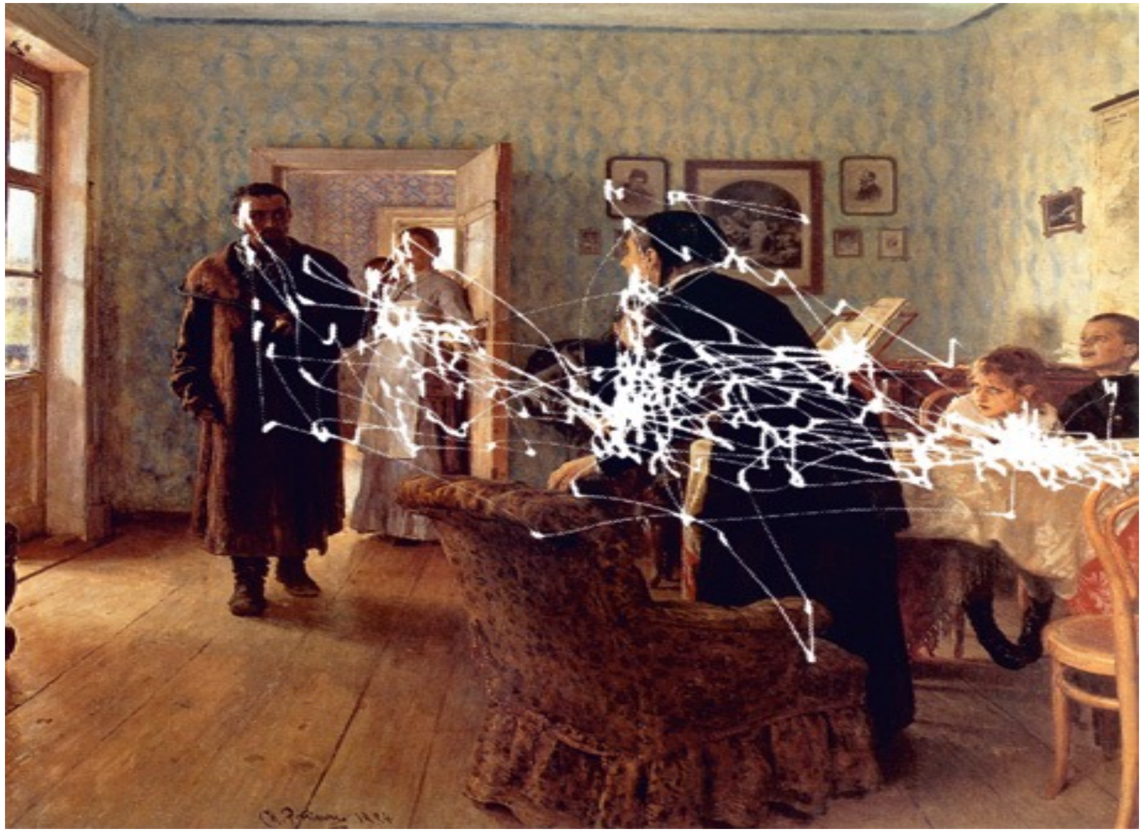


Ages of the People

# What directs our attention ?

Goal / Task demand

Clothes they are wearing

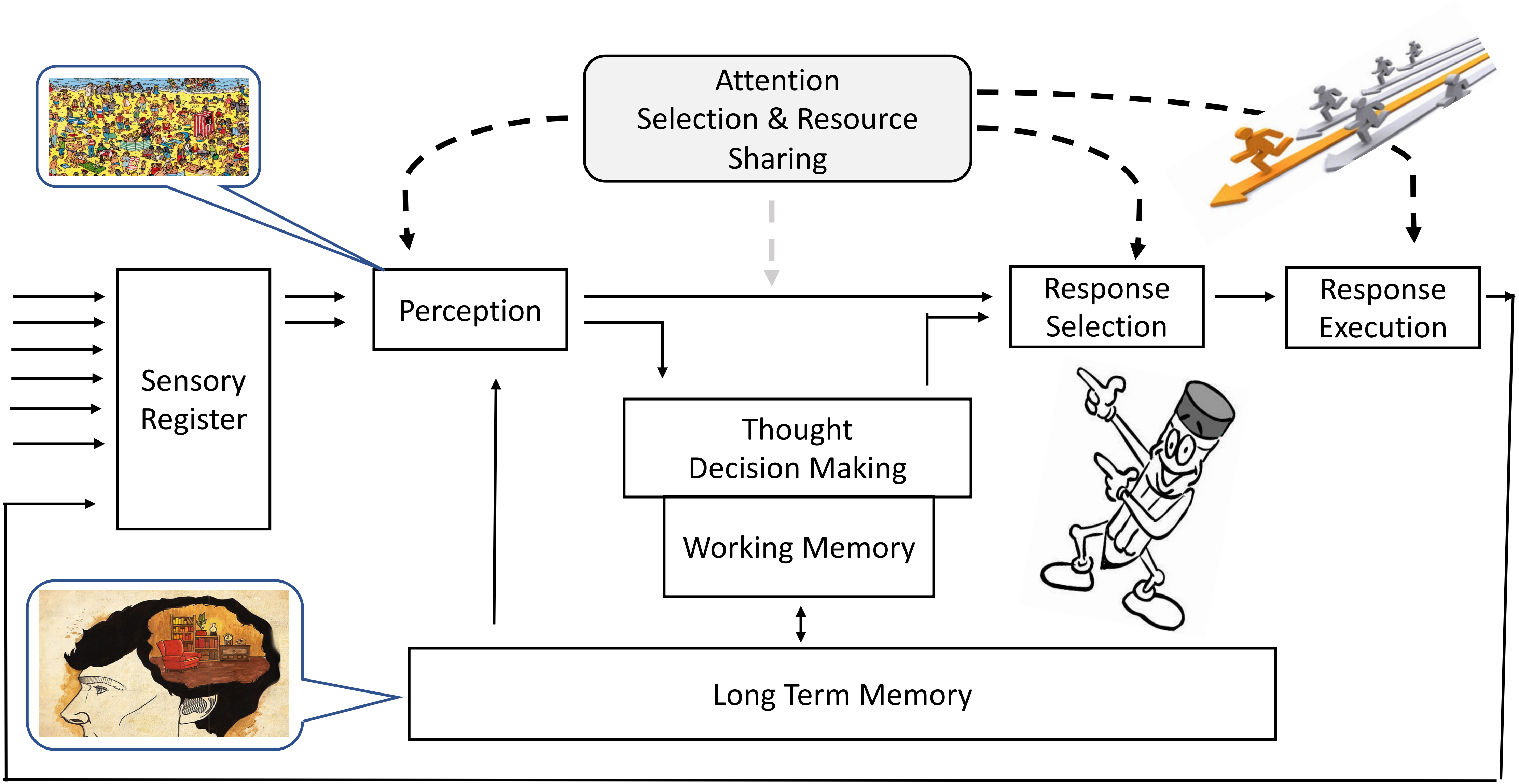


Activity prior to the visitor

# Where does attention take place?

- How the the cognitive system decides “where and when” to apply attentional filter ? Or How the system decides for early or late selection ?
- What happens to unattended information? To what extent we process them?
- Does the system differ across expertise?

# Loci of selection / Where does attention take place?



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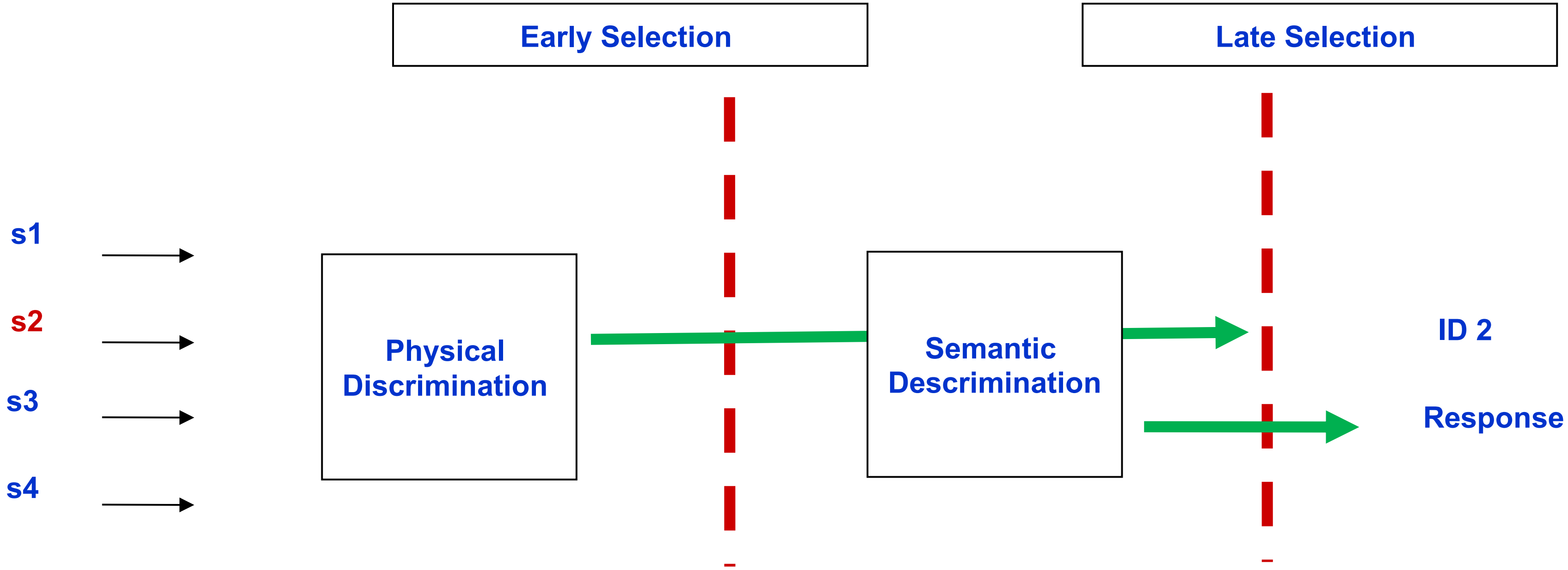



- While attended information is important for evaluation, understanding unattended information is also crucial
- Early refers to physical filtering whereas late selection model refers to semantic filtering

# Contd.

- How we test what is being attended?

- Dichotic listening task *(Cherry , 1953)*
  - Unattended stream – male/female voice, Loudness, strongly emotional words, taboo words
  - No meaningful recall of the unattended channel
  - Showed Early Selection

## Contd.

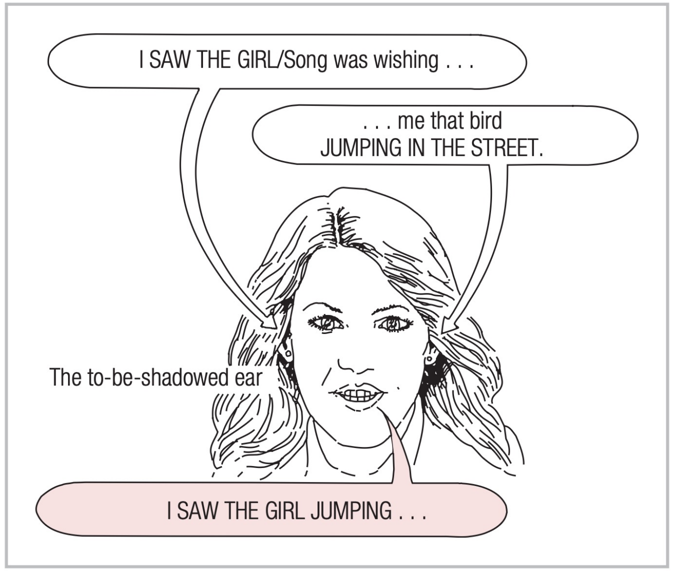
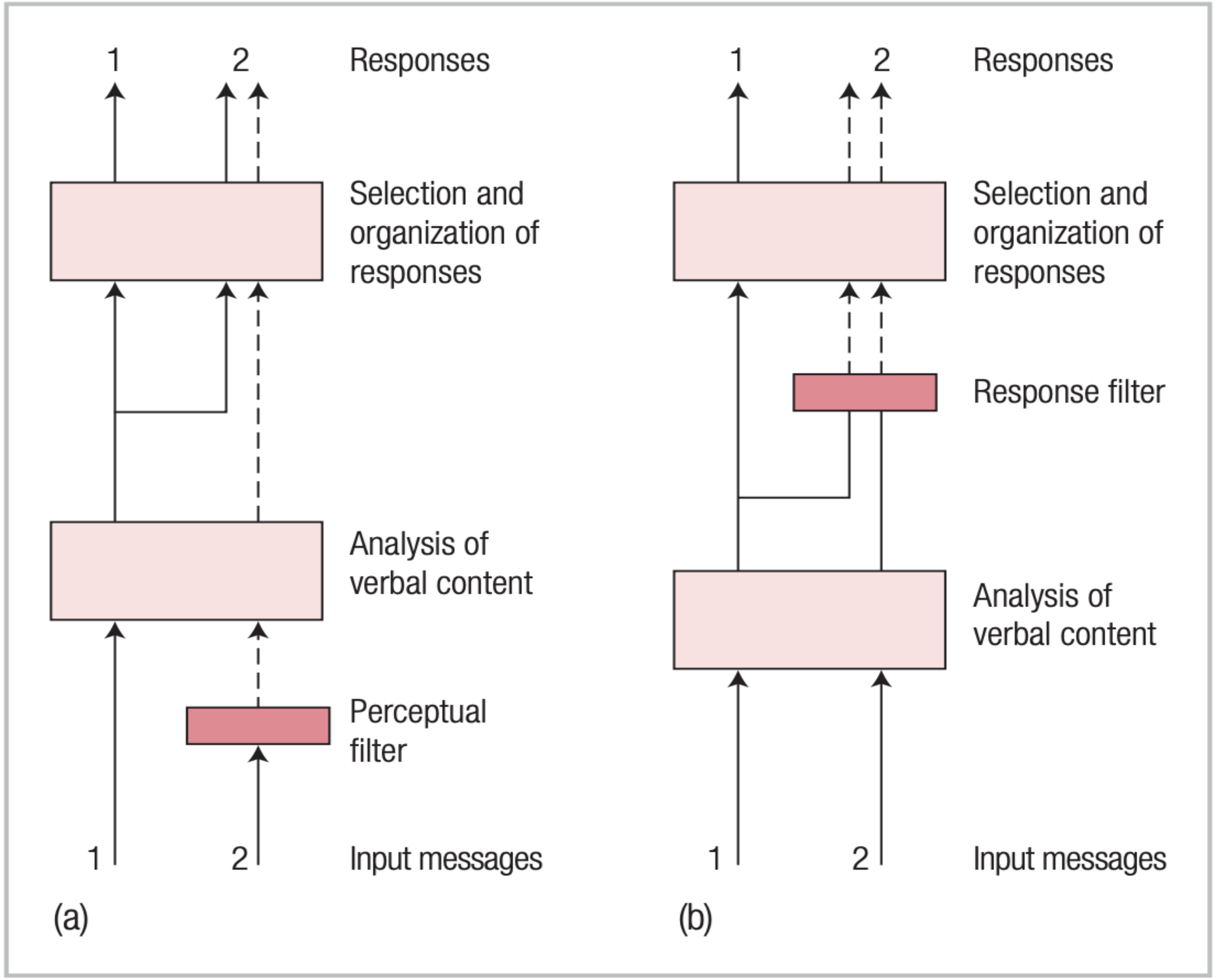



**FIGURE 3.4** An illustration of the Treisman (1960) experiment. The meaningful message moves to the other ear, and the participant sometimes continues to shadow it against instructions. (After Klatzky, 1975. Adapted by permission of the publisher. © 1975 by W. H. Freeman.)

Participant detected 87% of the target words in shadowed ear and only 8% in unshadowed ear.




a. Attenuation Model

b. Late Selection Model

**FIGURE 3.5** Treisman and Geffen's illustration of attentional limitations produced by (a) Treisman's (1964) attenuation theory and (b) Deutsch and Deutsch's (1963) late-selection theory. (From Treisman & Geffen, 1967. Reprinted by permission of the publisher. © 1967 by the *Quarterly Journal of Experimental Psychology.*)

**Dichotic listening** ***(McKay, 1973)***

- **Attended stream:** Ambiguous sentence

"They were throwing stones at the bank."

- **Unattended stream:** Biasing word

"River" or "Money"

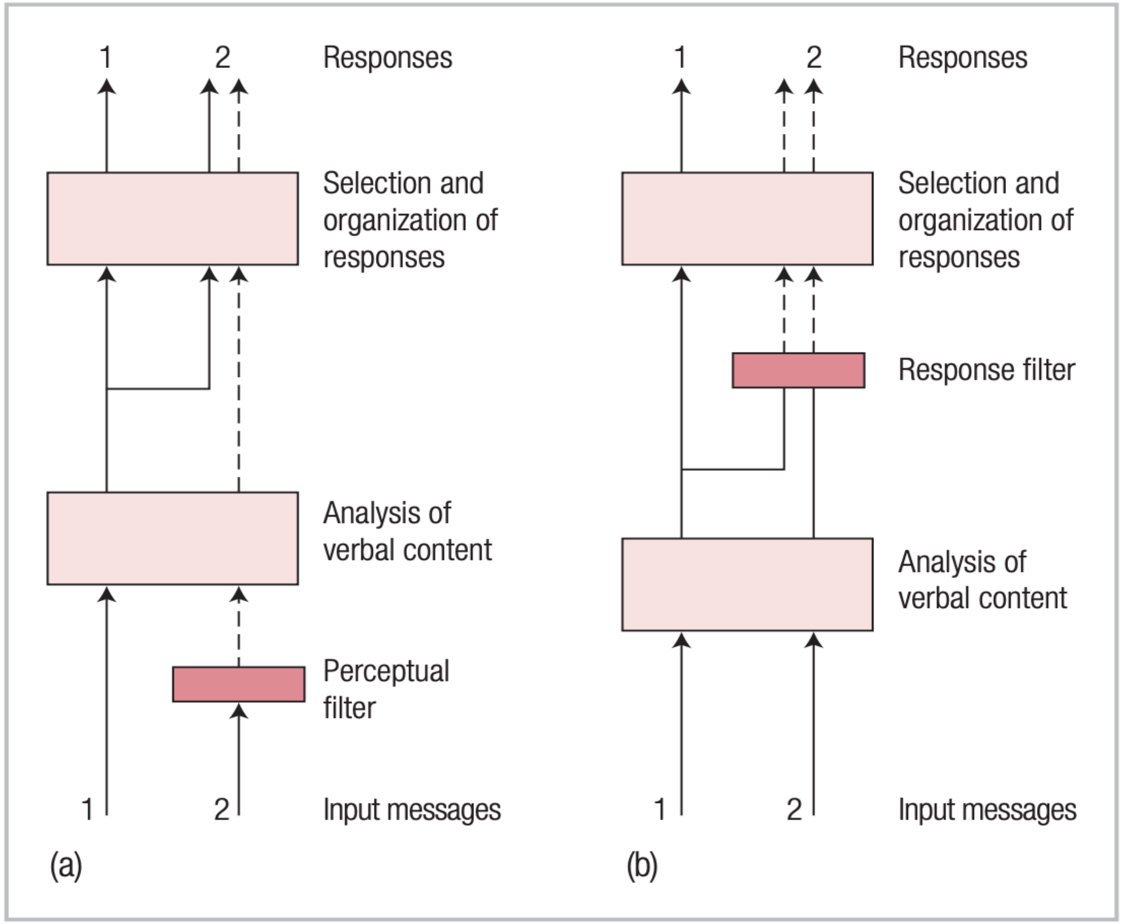
**The biasing word had an effect!**

If "money", the ambiguous sentence was more likely interpreted as financial institution

**Sentences to recognize :**

- They threw stones toward the side of the river yesterday
- They threw stones at the savings and loan association yesterday
- **Supporting late selection or attenuation view ?**

Contd.



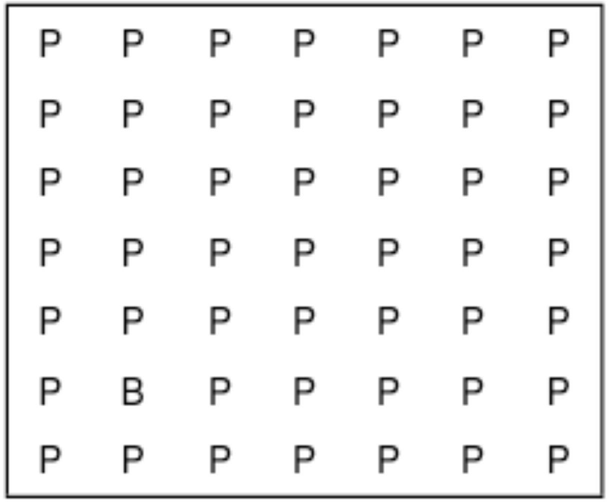

a. Attenuation Model    b. Late Selection Model

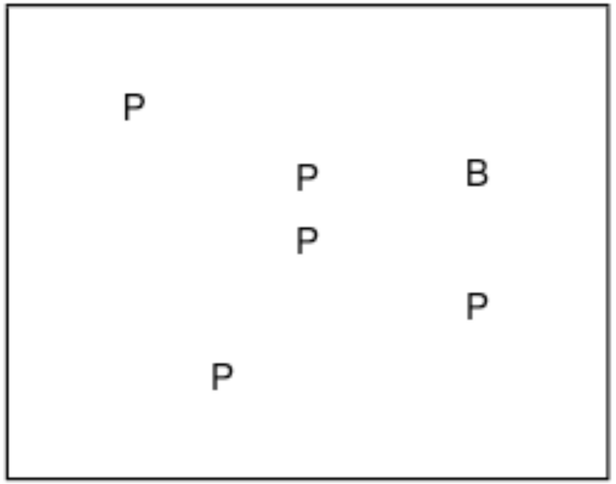
McKay (1973). Aspects of the theory of comprehension, memory, and attention. *Quarterly Journal of Experimental Psychology.*

## Contd.

- How much unattended information is processed?
- How much information is processed before attention?
- Early refers to physical filtering whereas late selection model refers to semantic filtering
- Both are “all or none” processing
- Triesman suggests about attenuation and function as per the task/ stimuli demands – flexible system

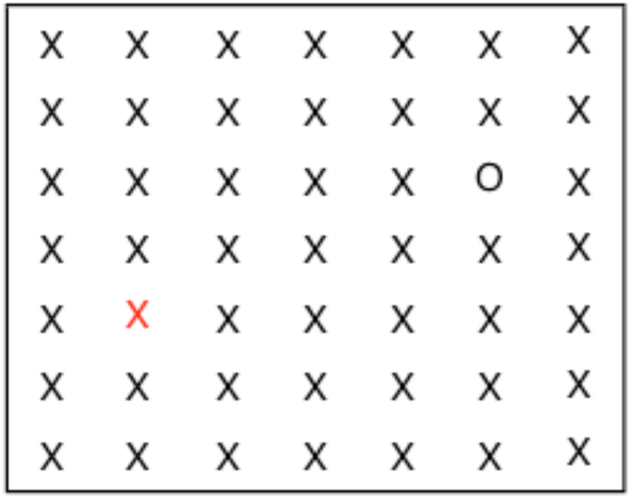
# Visual Search – Task

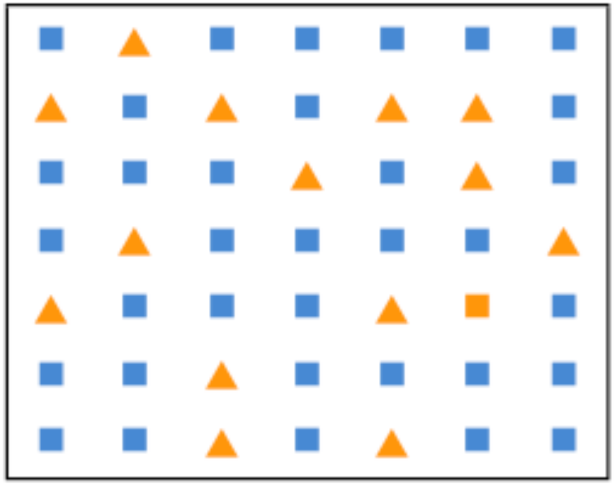



**RTs is a function of set size**

**Pop out effect**

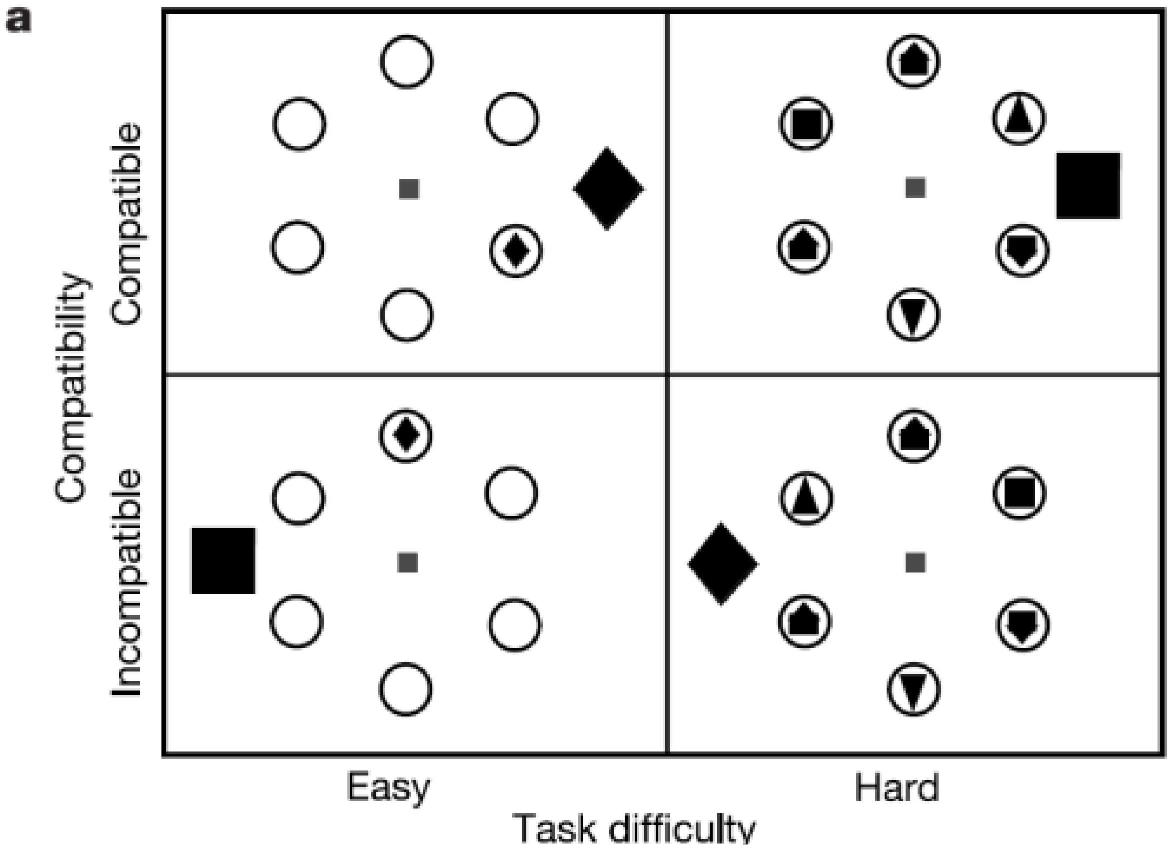


**Feature Search & Conjunction Search**

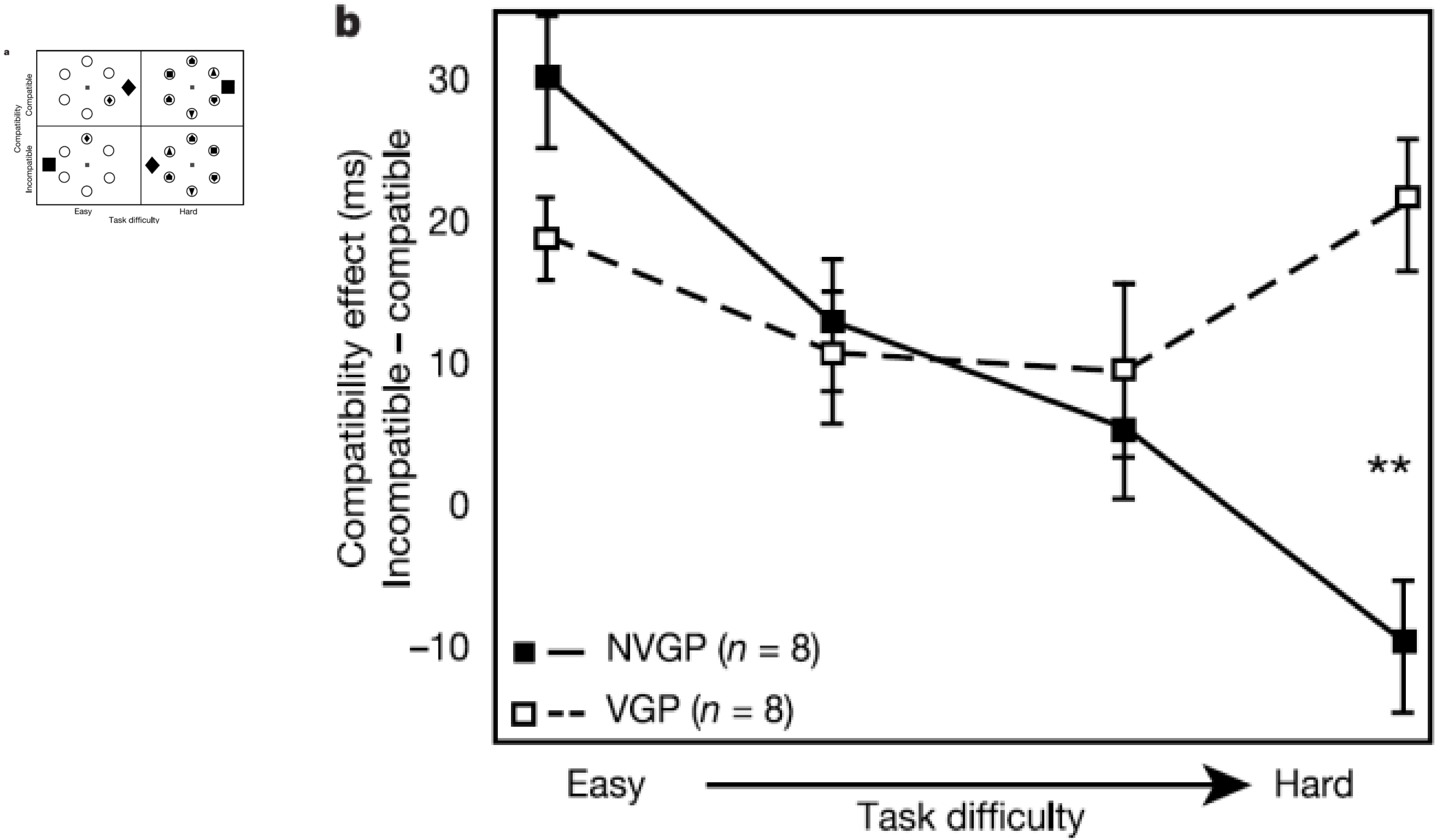
**How system decides for early or late selection ?**
**Does the system differ across expertise?**

- How experience and expertise modulates the <u>scope</u> of <u>attention</u>?
- How <u>perceptual complexity</u> determines the scope of attention?




Green and Bevelier, 2003

## How system decides for early or late selection ? Does the system differ across expertise?

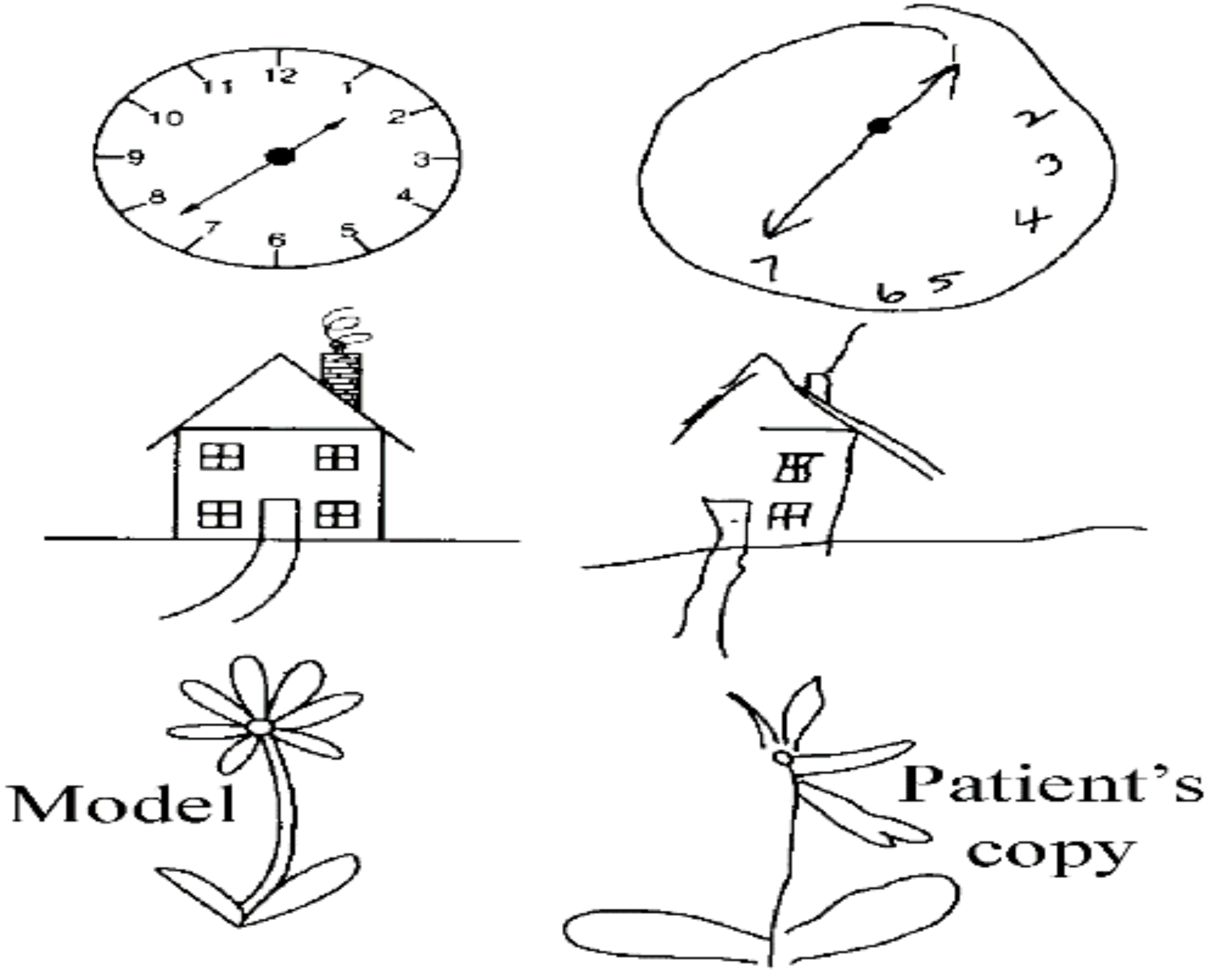


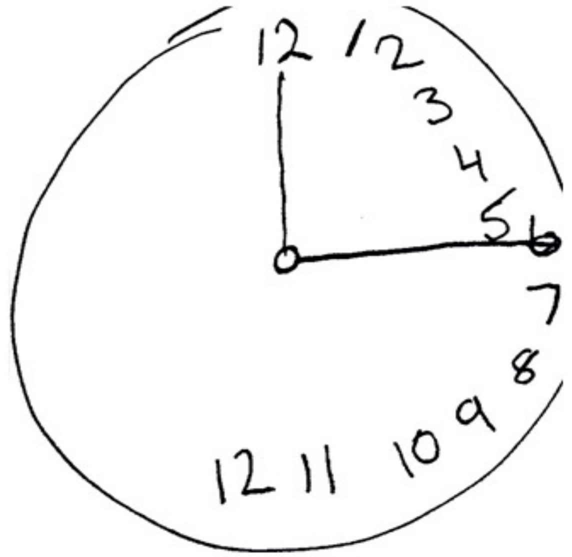

Green and Bevelier, 2003

## Attention deficits

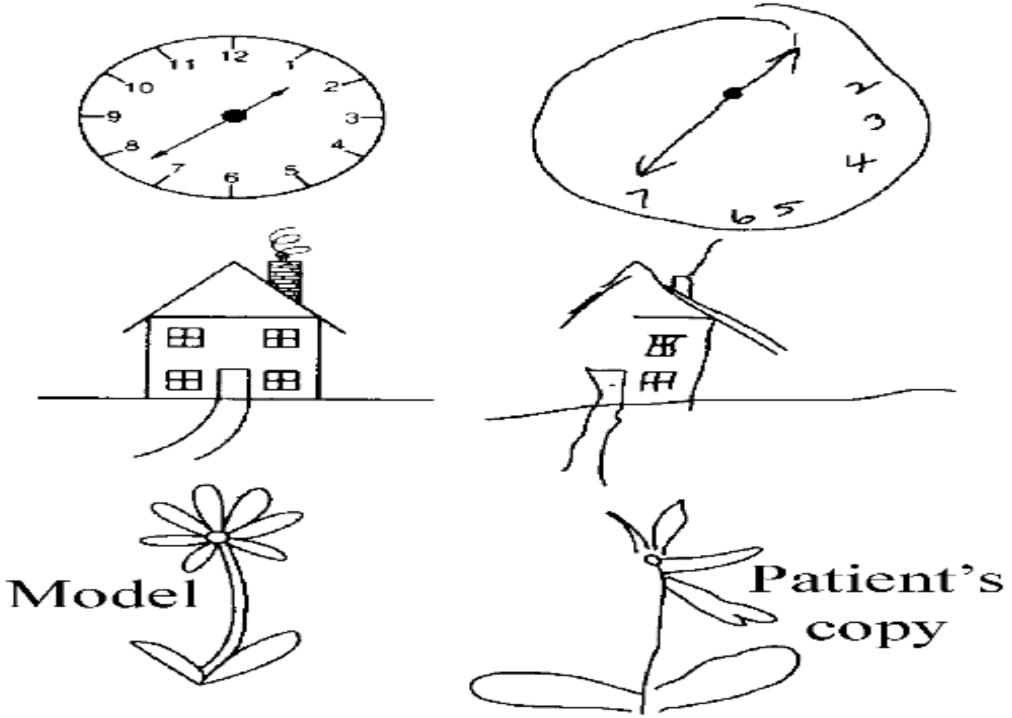
Difference between Perceptual, Memory, and Attention Problems
How Visual Neglect is not a Perceptual Problem and How Agnosia is not an Attention Problem

# Attention Problem - Visual Neglect / Hemi Neglect



# Visual Neglect / Hemi Neglect




- Patients with right parietal damage shows neglect to the left visual field and they ignore to even listen to the speaker from the left side, miss to eat or explore the left side if using touch.
- Patients with parietal lobe lesions have shown problem with imagery and their contralateral side memory performance – known as representational neglect

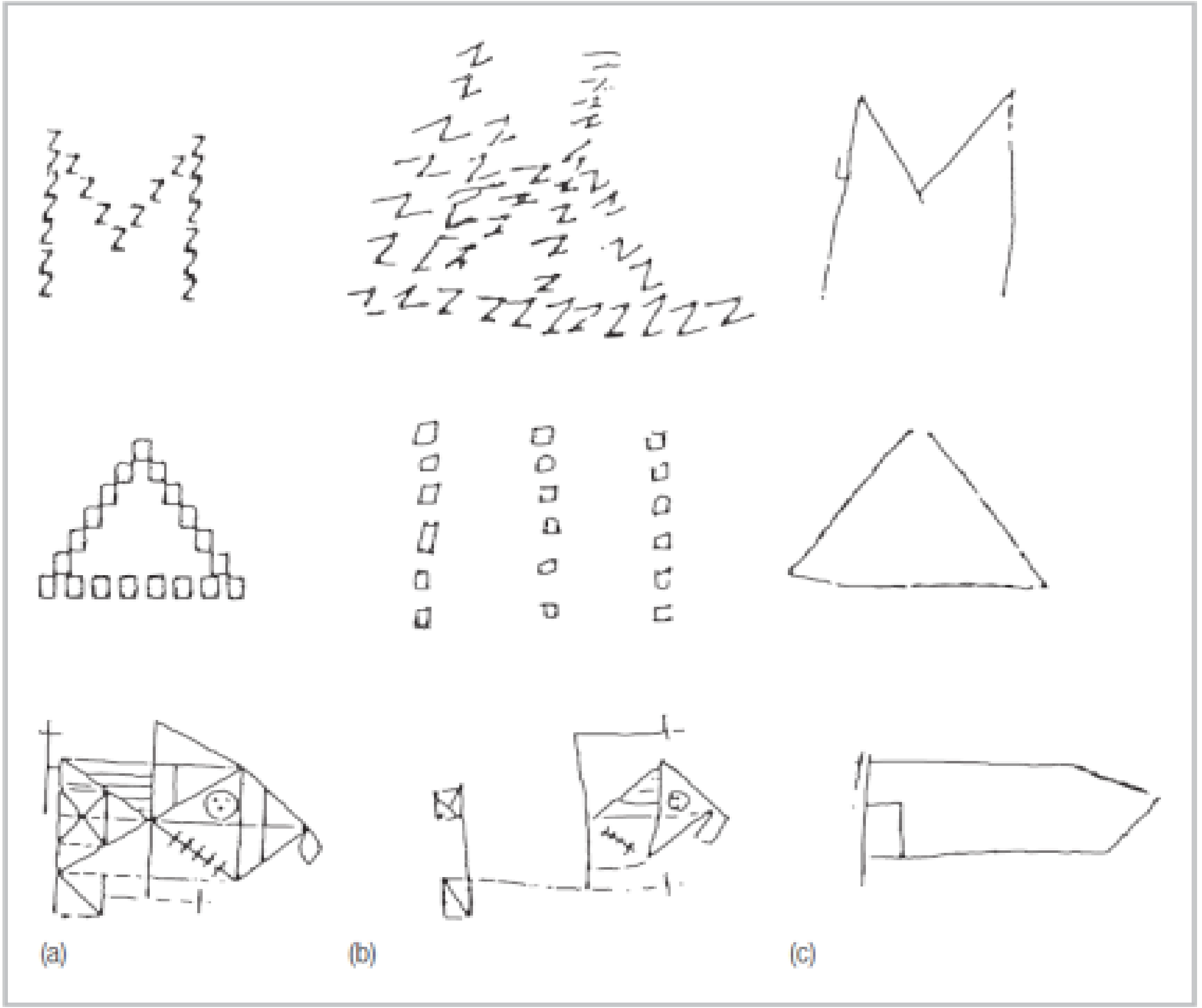
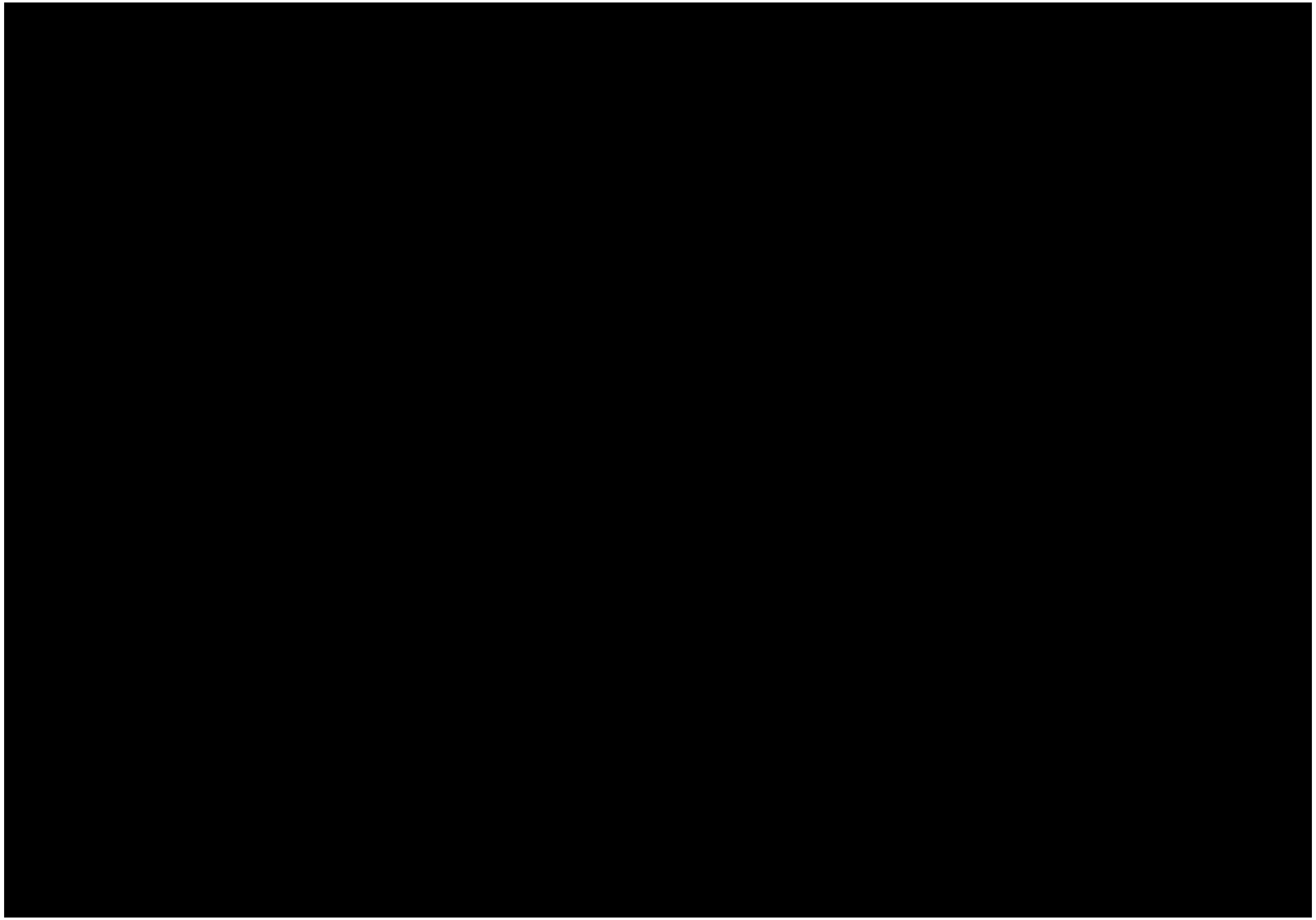
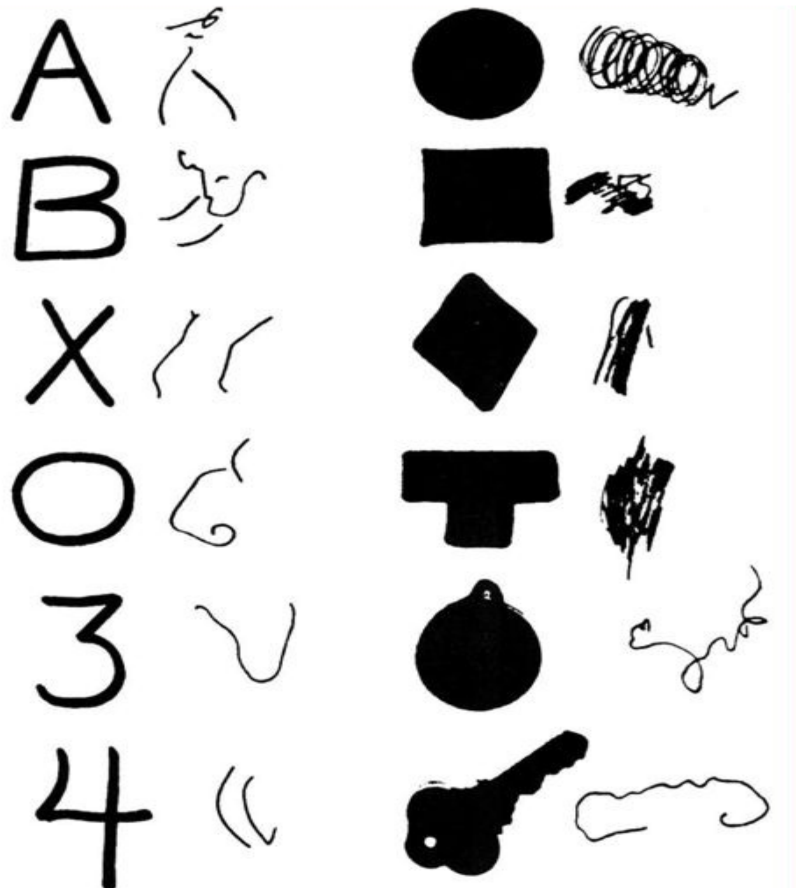



FIGURE 3.18 (a) The pictures presented to patients with parietal damage. (b) Examples of drawings made by patients with right-hemisphere damage. These patients could reproduce the specific components of the picture but not their spatial configuration. (c) Examples of drawings made by patients with left-hemisphere damage. These patients could reproduce the overall configuration but not the detail. (After Robertson & Lamb, 1991. Adapted by permission of the publisher. © 1991 by *Cognitive Psychology*.)

# Visual Neglect / Hemi Neglect

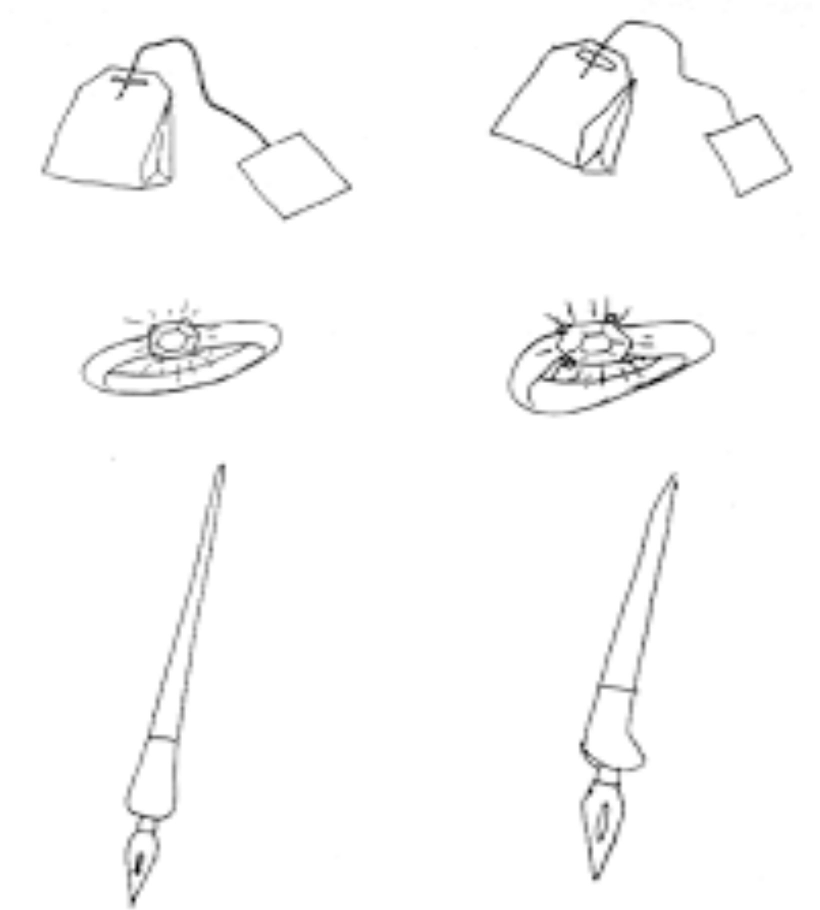


# Perceptual Problems - AGNOSIA




(Benson & Greenberg, 1969)

**Apperceptive – inability to recognize simple objects**



**Associative – inability to recognize complex objects**

# Summary

- Attention – Operational definition – focusing on selective nature
- Attention in relationship with Perception and Memory
- Factors that influence attentional processing
- Where does it take place in the information processing stream – determining the scope of attention
- Attention deficits

# Assignment

1. Observe yourself and one of your friends while reading online news across at least three news websites (e.g., BBC, The Guardian, The Wire, Newslaundry, or Times of India). Identify which website was easiest and most difficult to navigate, from main headlines to the specific news articles. Explain your observations using concepts of attention.

1. Further, choose at least one of the following factors: *knowledge, expectancy, or goal, and explain how it affects. visual* search while reading these news websites.